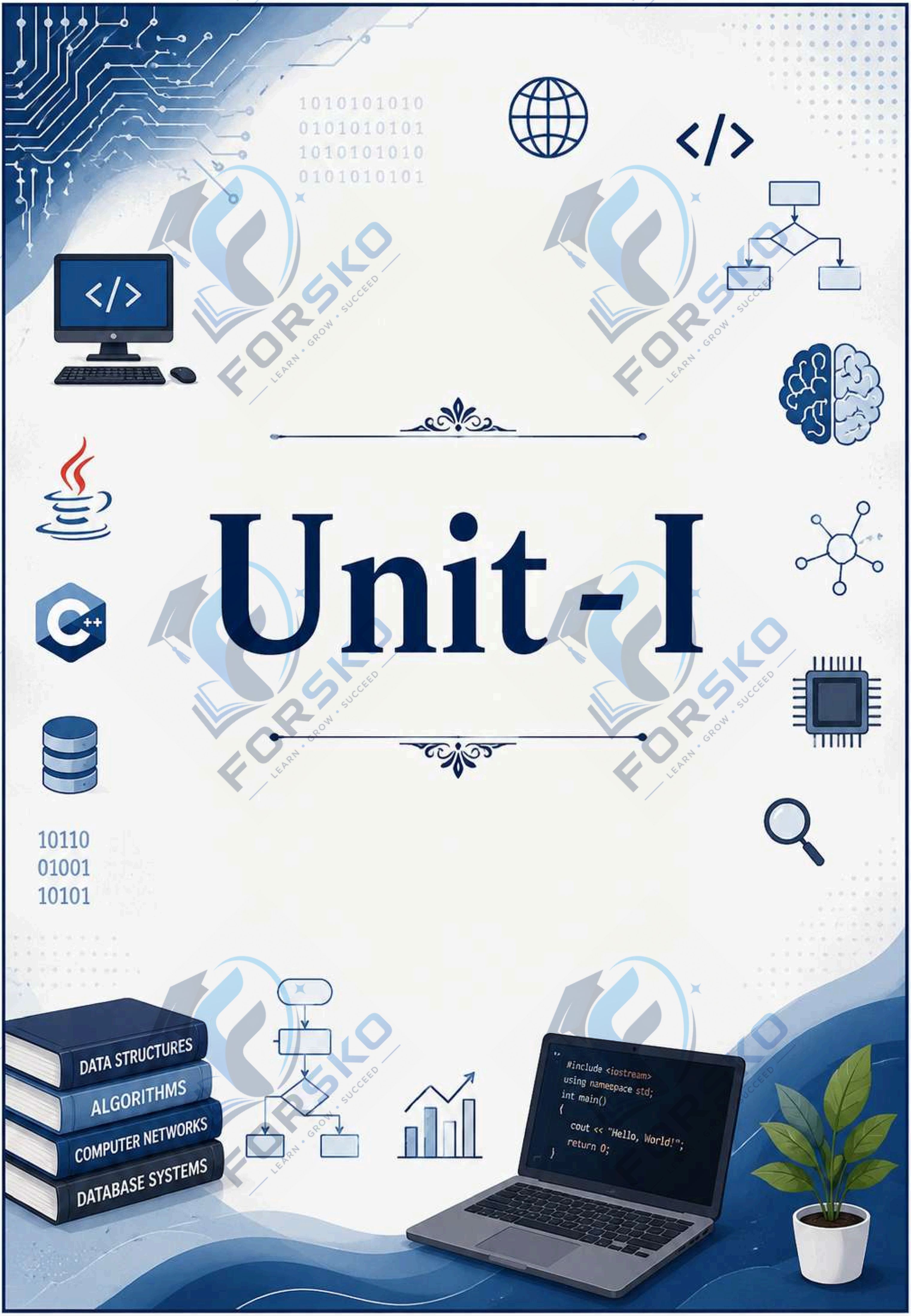
Unit-I



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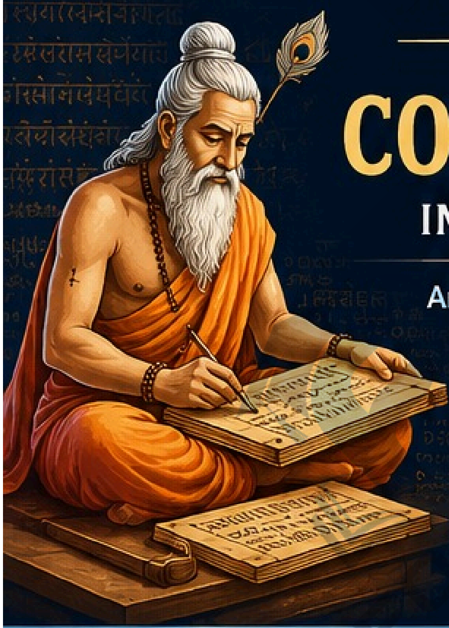
DATA STRUCTURES
ALGORITHMS
COMPUTER NETWORKS
DATABASE SYSTEMS
#include <iostream>
using namespace std;
int main()
{
cout << "Hello, World!";
return 0;
}



IKS IN COMPUTATIONAL SCIENCES

IMPORTANCE OF ANCIENT KNOWLEDGE

Ancient Wisdom • Timeless Logic • Modern Innovation

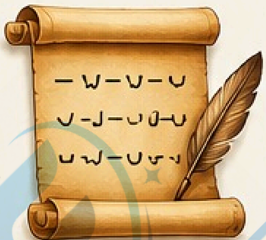
IKS (Indian Knowledge Systems) offers deep insights, logical frameworks and problem-solving approaches that align with the core of **Computational Sciences.**

ANCIENT ROOTS OF COMPUTATIONAL THINKING

PANINI'S ASHTADHYAYI

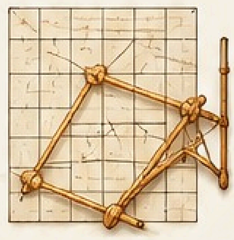
The grammar rules form a generative system, similar to modern formal languages and automata theory.

CHHANDAS (PROSODY)



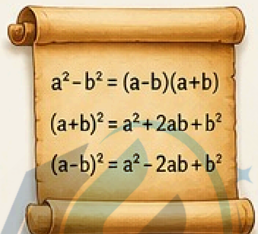
Binary patterns in Sanskrit meter resemble sequences and combinatorial structures used in coding theory.

SULBA SUTRAS (GEOMETRY)



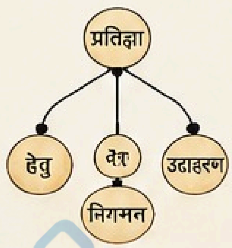
Algorithms for constructing altars involve precise geometric computations, similar to algorithmic problem solving.

VEDIC MATHEMATICS



Provides mental math techniques and fast algorithms still useful in optimization and complexity reduction.

NYAYA LOGIC



Logical reasoning frameworks in Nyaya are the foundation of inference systems and AI reasoning.

RELEVANCE IN COMPUTATIONAL SCIENCES

Algorithms & Problem Solving

Ancient methods in mathematics and logic inspire efficient algorithms.

Data Representation & Compression

Sanskrit grammar and prosody show advanced encoding concepts.

Formal Languages & Automata

Panini's grammar is a model for rule-based systems in CS.

Artificial Intelligence & Reasoning

Nyaya logic supports knowledge representation, expert systems and inference engines.

Optimization & Heuristics

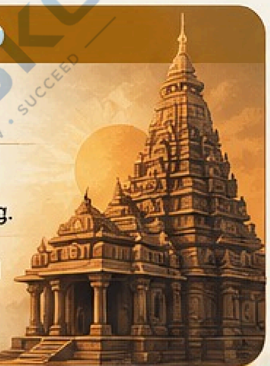
Vedic Mathematics provides fast methods for optimization and resource efficiency.

WHY IT MATTERS?

- IKS promotes a holistic and sustainable approach to computing.
- It strengthens algorithmic thinking with time-tested wisdom.
- It bridges ethics, logic and innovation.
- It gives India a unique perspective in global technology.

THE WAY FORWARD

- Integrate IKS concepts in CS curriculum.
- Encourage research at the intersection of ancient knowledge and modern computing.
- Preserve and propagate India's intellectual heritage for future innovations.



From Rishis to Algorithms – A Legacy of Knowledge, Logic & Infinity

ॐ

Rooted in IKS, Rising in Innovation.

DEFINING INDIAN KNOWLEDGE SYSTEM

A Timeless Heritage. A Sustainable Future.

WHAT IS INDIAN KNOWLEDGE SYSTEM (IKS)?

Indian Knowledge System (IKS) is the collective knowledge, wisdom and practices developed in India over thousands of years. It is holistic, value-based and experience-driven knowledge that encompasses the understanding of the self, society, nature and the universe.

IKS

- Holistic
- Value-based
- Experience-driven
- Universal
- Timeless

KEY CHARACTERISTICS

- Rooted in ancient wisdom yet relevant today.
- Promotes harmony between individual, society and nature.
- Transmitted through oral traditions, texts, practices and lived experiences.
- Interdisciplinary and interconnected in its approach.
- Aims for the well-being of all (Sarve Bhavantu Sukhinah).

CORE PILLARS OF IKS

DHARMA
The foundation of righteous living, ethics and duties.

JNANA
Pursuit of knowledge through study, reflection and wisdom.

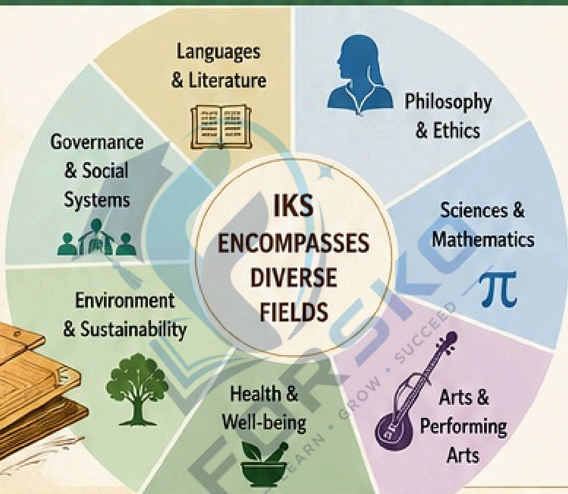
KARMA
Action with responsibility and selflessness for the welfare of all.

YOGA
Union of body, mind and spirit for inner and outer balance.

VASUDHAIVA KUTUMBAKAM
The world is one family promoting unity and compassion.

RITA
The cosmic order that sustains life and harmony.

DOMAINS OF INDIAN KNOWLEDGE SYSTEM

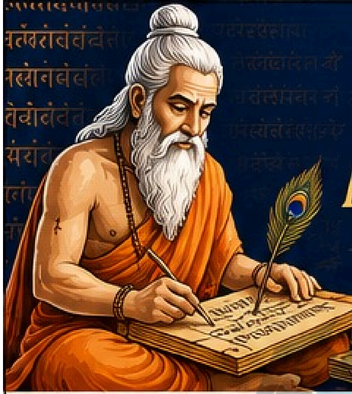


WHY IKS MATTERS TODAY?

- Provides sustainable solutions to modern challenges.
- Enhances mental well-being and holistic development.
- Inspires innovation through traditional wisdom and practices.
- Promotes cultural identity and global harmony.
- Supports inclusive and value-based education and research.

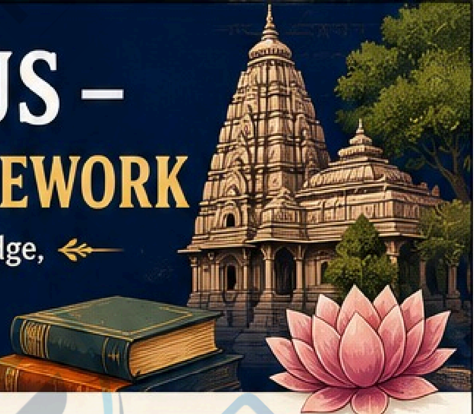
"From the wisdom of the past, for the innovation of the future.
IKS – Our Legacy, Our Strength, Our Future."

Ancient in Origin | Timeless in Wisdom | Relevant in Every Age | For a Better Tomorrow



THE IKS CORPUS – A CLASSIFICATION FRAMEWORK

Systematizing Ancient Wisdom for Knowledge, Research and Innovation



WHAT IS THE IKS CORPUS?

The **IKS Corpus** is a structured collection of knowledge resources derived from India's ancient wisdom traditions. This classification framework organizes diverse knowledge systems into coherent categories to facilitate understanding, accessibility, research and application in modern contexts.

- Preserves Civilizational Wisdom
- Enables Interdisciplinary Research
- Promotes Sustainable Solutions
- Empowers Future Generations

THE IKS CORPUS CLASSIFICATION FRAMEWORK

1 SOURCE LITERATURE

Primary texts and original scriptures that form the foundation of IKS.

Examples

- Vedas & Upanishads
- Itihasas & Puranas
- Agamas & Tantras
- Sutras & Samhitas

2 KNOWLEDGE DOMAINS

Broad knowledge areas that encompass the various dimensions of Indian wisdom.

Examples

- Dharmic Sciences
- Life Sciences (Ayurveda)
- Mathematics & Logic
- Arts, Aesthetics & Culture
- Ecology & Environment
- Governance & Ethics

3 SUB-DOMAINS & DISCIPLINES

Specific disciplines within each domain with distinct bodies of knowledge.

Examples

- Ayurveda (under Life Sciences)
- Vyakarana (under Linguistics)
- Shilpa Shastra (under Arts & Architecture)

4 CONCEPTS & PRINCIPLES

Fundamental ideas, theories and principles that explain natural, social and spiritual phenomena.

Examples

- Panchamahabhutas
- Karma & Dharma
- Rta (Cosmic Order)
- Triguna Theory

5 PRACTICES & APPLICATIONS

Methods, techniques and systems applied in daily life, professions and societal well-being.

Examples

- Yoga & Meditation
- Agriculture Practices
- Architectural Techniques
- Medical Therapies
- Conflict Resolution Models

6 TOOLS, SYSTEMS & TECHNOLOGIES

Indigenous tools, systems and technologies developed through traditional knowledge.

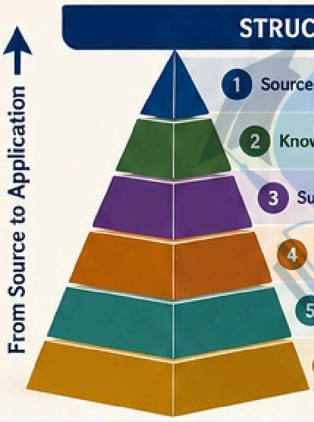
Examples

- Water Management Systems
- Temple Architecture
- Metallurgy (Wootz Steel)
- Astronomical Instruments

CROSS-CUTTING PERSPECTIVES (APPLICABLE ACROSS ALL CATEGORIES)

- Sustainability
- Holistic & Integral View of Life
- Ethics & Righteousness
- Interconnectedness of All Beings
- Spiritual & Consciousness Dimensions

STRUCTURE AT A GLANCE



1	Source Literature	Original texts and scriptures
2	Knowledge Domains	Broad areas of wisdom
3	Sub-domains & Disciplines	Specific fields of knowledge
4	Concepts & Principles	Core ideas and theories
5	Practices & Applications	Methods and real-world use
6	Tools, Systems & Technologies	Indigenous innovations and systems

WHY THIS FRAMEWORK MATTERS

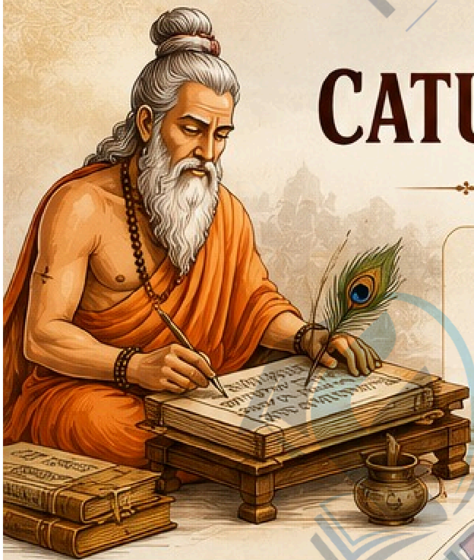
- Organizes vast and diverse knowledge in a systematic manner.
- Enhances discoverability, accessibility and usability.
- Supports interdisciplinary research and innovation.
- Bridges ancient wisdom with modern challenges.
- Strengthens cultural identity and knowledge sovereignty.

“IKS is not just knowledge of the past, but a living wisdom that can light the path to a harmonious and sustainable future.”

- Preserve our Heritage
- Advance Knowledge
- Serve Humanity
- Inspire the World

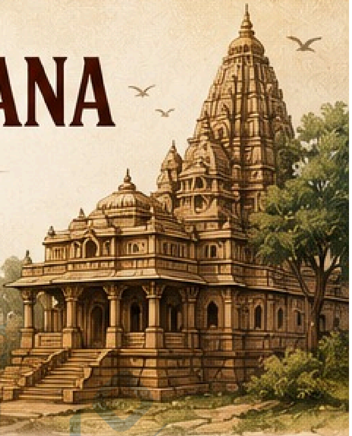
|| लोकाः समस्ताः सुखिनो भवन्तु ||

May all beings be happy and free.



CATURDASA – VIDYASTHANA

The Fourteen Abodes of Knowledge



Caturdasa – Vidyasthana refers to the fourteen traditional fields of knowledge described in Indian Knowledge Systems (IKS).
They represent a holistic vision of learning that nurtures the body, mind, intellect and spirit.

“यत्र विश्वं भवत्येकनीडम्
— ज्ञानं परम् बलम्

Where the world becomes one nest – Knowledge is the highest strength.

ROOTS IN TRADITION

These 14 vidyasthanas are mentioned in ancient texts like Kāśyapa Śikṣā, purāṇas and smṛtis. They were the foundation of education in ancient India.

THE 14 VIDYASTHANA – CATEGORIES

VICHARA (SACRED KNOWLEDGE)

1. Ṛg Veda
2. Yajur Veda
3. Sāma Veda
4. Atharva Veda

The Four Vedas – foundation of dharma, rituals, wisdom and spiritual understanding.

VEDĀNGA (AUXILIARY DISCIPLINES)

5. Śikṣā (Phonetics)
6. Vyākaraṇa (Grammar)
7. Nirukta (Etymology)
8. Chanda (Metrics)
9. Jyotiṣa (Astronomy)

Supporting sciences that help in understanding and preserving the Vedic wisdom.

UPANGA (APPLIED DISCIPLINES)

10. Kalpa (Rituals & Procedures)
11. Śāstra (Logic & Knowledge Systems)
12. Darśana (Philosophy)

Practical and philosophical knowledge for living a meaningful and righteous life.

JĪVIKĀ (LIFE APPLICATIONS)

13. Āyurveda (Medical Science)
14. Dhanurveda (Martial Science)

Knowledge for health, protection and overall well-being of individual and society.

SIGNIFICANCE

- Promotes holistic development of body, mind and spirit.
- Integrates spiritual, intellectual and practical knowledge.
- Encourages harmony between individuals, society and nature.
- Forms the base of India's rich cultural and scientific heritage.
- Still relevant for modern education, research and innovation.

“सा विद्या या विमुक्तये”

“That is knowledge which liberates.”

TIMELESS RELEVANCE

Spiritual Wisdom | Scientific Thinking | Ethical Living | Sustainable Society

The Caturdasa – Vidyasthana shows that Indian Knowledge Systems are universal, integrated and eternally relevant.

One Knowledge – Many Dimensions. One Goal – Human Excellence.

HISTORY OF IKS

The Timeless Journey of INDIAN KNOWLEDGE SYSTEMS

"IKS is the collective wisdom, knowledge and practices developed in India over thousands of years. It reflects a holistic understanding of life, nature and the universe."

IKS – OUR HERITAGE, OUR STRENGTH

- Ancient in Origin
- Timeless in Wisdom
- Holistic in Approach
- Universal in Relevance
- Sustainable in Nature
- Innovative in Spirit

JOURNEY THROUGH TIME

1. VEDIC PERIOD (1500 – 600 BCE)

ऋग्वेद

Knowledge preserved in oral traditions and scriptures like the Vedas. Focus on rituals, philosophy, nature worship and holistic living.

2. EPIC & SHASTRIC PERIOD (600 BCE – 200 CE)

उपनिषद, महाभारत, रामायण

Expansion of knowledge through Upanishads, Itihasas, Puranas and Shastras. Growth of dharma, ethics, polity, Ayurveda, grammar, architecture and arts.

3. CLASSICAL AGE (200 CE – 1200 CE)

Flourishing of universities (Nalanda, Takshashila, Vikramashila). Advancements in mathematics, astronomy, medicine, metallurgy, logic, literature and performing arts.

4. MEDIEVAL PERIOD (1200 CE – 1750 CE)

Knowledge continued in gurukuls, mathas, temples and craft guilds. Synthesis of indigenous knowledge with new ideas in science, arts, language, spirituality and society.

5. COLONIAL PERIOD (1750 CE – 1947 CE)

IKS faced challenges due to colonial education systems. Many knowledge traditions were underrepresented, yet preserved by scholars, saints and communities.

6. MODERN ERA & BEYOND (1947 – PRESENT)

Revival, research and integration of IKS in education, science, technology, policy and daily life for a sustainable and inclusive future.

CORE DIMENSIONS OF IKS

- **Spiritual & Philosophical** – Explores consciousness, self-realization and the purpose of life.
- **Scientific & Rational** – Based on observation, experimentation and logical reasoning.
- **Ethical & Values Based** – Promotes dharma, compassion, harmony and responsibility.
- **Practical & Experiential** – Learning through experience, practice and application.
- **Sustainable & Ecological** – Lives in balance with nature and supports sustainability.
- **Artistic & Cultural** – Nurtures creativity, beauty, aesthetics and expression.

EXAMPLES OF IKS CONTRIBUTIONS

- **Mathematics** – Zero, Decimal System, Algebra, Trigonometry
- **Astronomy** – Planetary Movements, Eclipses, Calendars
- **Medicine (Ayurveda)** – Holistic health, herbs, wellness, yoga
- **Architecture** – Vastu Shastra, temple architecture, urban planning
- **Metallurgy** – Iron pillar of Delhi, Zinc distillation
- **Agriculture** – Crop rotation, irrigation, soil management
- **Language & Literature** – Sanskrit grammar, classical literature, poetry, drama
- **Arts & Performing** – Music, dance, painting, sculpture, theatre

WHY KNOWING OUR HISTORY MATTERS?

- Builds pride in our rich intellectual heritage.
- Strengthens cultural identity and continuity.
- Inspires innovation by learning from the past.
- Supports sustainable and inclusive development.
- Shares India's wisdom for global well-being.

"Take up one idea. Make that one idea your life – think of it, dream of it, live on that idea. Let the brain, muscles, nerve, every part of your body be full of that idea, and just leave every other idea alone. This is the way to success."

– Swami Vivekananda

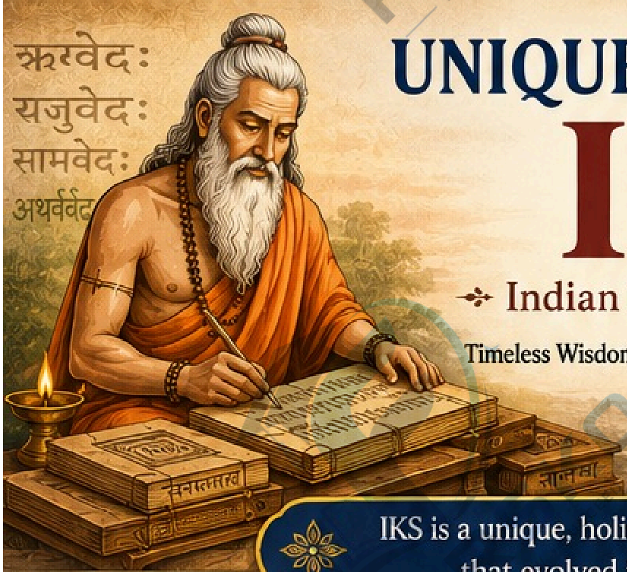
IKS is not just history – it is a living tradition. By understanding our past, we can create a better, wiser and compassionate future.

Learn our roots · Live our values · Lead our future

"From eternal wisdom to endless possibilities."

IKS – Past, Present & Future United.

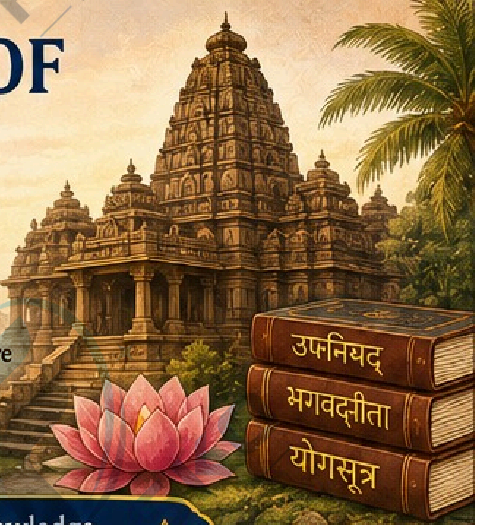
"सा विद्या या विमुक्तये" – That is knowledge which liberates.



UNIQUE ASPECTS OF IKS

Indian Knowledge Systems

Timeless Wisdom • Holistic Vision • Sustainable Future



IKS is a unique, holistic and time-tested body of knowledge that evolved in India over thousands of years.

Holistic

IKS integrates physical, mental, intellectual, emotional and spiritual well-being to achieve complete human development.

Value-Based

Rooted in ethics, dharma and moral principles, IKS promotes righteous living, responsibility and harmony in society and nature.

Interdisciplinary

IKS encompasses multiple disciplines including philosophy, science, medicine, arts, engineering, astronomy, governance and more.

Timeless

IKS has been continuously practiced and refined for thousands of years, demonstrating its enduring relevance.

Sustainable

IKS advocates living in harmony with nature, promoting sustainable lifestyles, conservation and responsible use of resources.

Innovative

IKS encourages curiosity, creativity and exploration, leading to innovative solutions for contemporary challenges.

Contextual

IKS evolved from India's unique cultural, social and geographical context, making it deeply relevant to Indian society.

Ancient Yet Relevant

While ancient in origin, the principles of IKS remain highly relevant for addressing modern world problems.

People-Centric

IKS places the individual and society at the center, aiming for the well-being of all (Sarve Bhavantu Sukhinah).

Global Appeal

The universal values of IKS such as peace, harmony, compassion and sustainability have global relevance and applicability.

IKS is not just a legacy of the past, but a guiding light for a knowledge-driven, value-based and sustainable future.

From Vedas to Modern Innovation

From Self to Society

From Nature to Sustainability

From India to the World

वसुधैव कुटुम्बकम्

One Earth • One Family • One Future

COMPONENTS OF A LANGUAGE

Language is a system of communication made up of interconnected components that work together to convey meaning.

THE MAIN COMPONENTS

1 PHONOLOGY

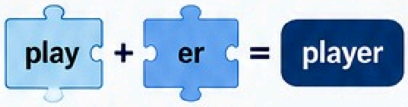
The system of speech sounds in a language and the rules for their combination.

Example:

Sounds: /p/ /a/ /t/
Combine to form: "pat"

2 MORPHOLOGY

The study of the structure of words and how meaningful units (morphemes) form words.



Example:

Root: play
Morpheme: -er (one who)
Word formed: player

3 SYNTAX

The set of rules that govern the arrangement of words to form phrases, clauses and sentences.

The | student | reads | a | book.

Example:

Correct Order = Meaning
Change in order can change the meaning.

4 SEMANTICS

The study of meaning in language – the meanings of words, phrases and sentences.

Example:

The word "tree" refers to a large plant with a trunk, branches and leaves.

5 PRAGMATICS

The study of how context influences meaning in communication.

Example:

The same sentence can be a request, an offer or a question depending on the context.

6 LEXICON

The vocabulary of a language – the words and their meanings stored in a speaker's mind.

Example:

Words like: run, happy, quickly, beautiful, computer, knowledge.

7 DISCOURSE

The use of language beyond the sentence level to create coherent and meaningful spoken or written texts.

Example:

A paragraph, story or conversation that has a beginning, middle and end.

8 GRAPHOLOGY

The system of writing – the symbols and rules used to represent language in written form.

Example:

Alphabets, scripts (Devanagari, Latin, Tamil, etc.) and punctuation used in writing.

9 SOCIO-LINGUISTICS

The study of the relationship between language and society, including dialects, registers, varieties and language change.

Example:

Different languages, accents and styles used in different social situations.

HOW THEY WORK TOGETHER

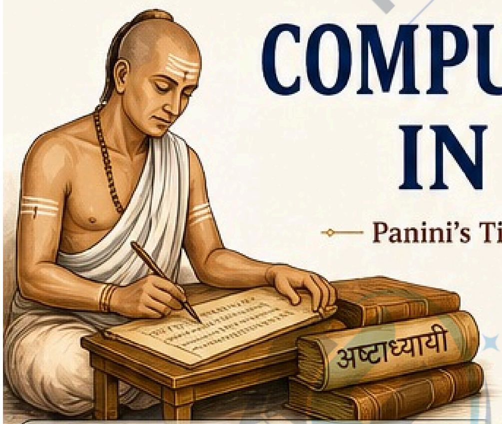
All components interact to help us produce, understand and interpret meaningful communication.

Sounds (Phonology) → Build Words (Morphology) → Form Sentences (Syntax) → Convey Meaning (Semantics) → Use in Context (Pragmatics) → Create Discourse (Discourse) → Write (Graphology) → Vary with Society (Socio-linguistics)

IN A NUTSHELL: Language is a complex system made of many parts.
Understanding these components helps us understand how human communication works so beautifully!

Unit - II

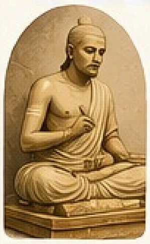
COMPUTATIONAL CONCEPT IN AṢṬĀDṚHYĀYĪ



Panini's Timeless Grammar – A Masterpiece of Computation

"Aṣṭādhyāyī is not just a grammar, it is an algorithmic system that generates infinite correct sentences from finite rules."

WHAT IS AṢṬĀDṚHYĀYĪ?



A systematic and generative grammar of Sanskrit composed by Maharshi Pāṇini. It precisely describes how words are formed and combined using a set of compact rules (sūtras). It is a perfect example of a formal system – like a computer program.

WHY IS IT COMPUTATIONAL?

- Rule-based System
- Generative Power
- Handles Recursion
- Highly Structured & Compact

AṢṬĀDHYĀYĪ AS AN ALGORITHM

Input (Dhātu / Prātipadika) → Apply Rules (Sūtras) → Output (Word / Sentence)

Recursive Application of Rules

KEY COMPUTATIONAL CONCEPTS IN AṢṬĀDḤYĀYĪ

1. RULE BASED SYSTEM

Pāṇini uses thousands of sūtras (precise rules) to generate words and sentences.

Example (1.3.2)
हलन्त्यम्
(halantyam)
– The last letter is a consonant.

Like instructions in a computer program.

2. FORMAL LANGUAGE

Aṣṭādhyāyī is a formal system with a well-defined syntax (sūtra structure) and semantics (meaning).

Sūtra Structure:
प्रत्याहार + अधिकार + विधि
(Pratyāhāra + Adhikāra + Vidhi)

Like the syntax of a programming language.

3. META RULES AND OPERATORS

Use of markers like अधिकार (scope), अनुवृत्ति (ellipsis), इट् (markers) to control rule application.

Example:
Anuvṛtti (eillipsis) reduces repetition
→ Saves space & time

Like control statements and operators.

4. RECURSION

Rules can be applied repeatedly until no more rules are applicable.

Example:
धातु → प्रातिपदिक → सुप्/तिङ् प्रत्यय → शब्द → वाक्य

Like recursive functions in programming.

5. COMPACTNESS & OPTIMIZATION

Pāṇini achieves maximum output with minimum input (few words – maximum coverage).

A highly optimized system – like efficient code.

6. ABSTRACT REPRESENTATION (PRATYĀHĀRA SYSTEM)

Pāṇini uses pratyāhāras (e.g., अच्, हल्) to represent large sets of sounds with short symbols.

Example: अच् = अ, इ, उ, ऋ, लृ, ए, ओ, औ ...

Like using variables or arrays.

7. CONDITIONAL APPLICATION (ADHIKĀRA)

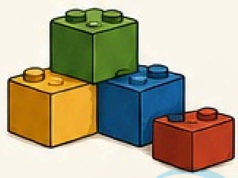
Rules are applied only in a particular context or domain.

Example (2.3.1)
सुपां सुलुक्
(supāṃ suluk)
– The affix 'su' is deleted.

Like if-else conditions.

8. MODULAR DESIGN

The grammar is divided into 8 chapters, 4 pādas each, with logical grouping.



Like modular programming.

9. DETERMINISM & NON-AMBIGUITY

Every valid input has a unique output. The system removes ambiguity using rule priority (परिभाषा rules).

Deterministic algorithm.

EXAMPLE: DERIVATION OF THE WORD "गच्छति" (gacchati)

Step	Operation	Sūtra Used (Example)	Result
1	Take root (धातु)	गम् (gam)	गम्
2	Add लट् लकार (present tense)	लट् (lat) – 3rd person singular	गम् + लट्
3	Add तिङ् प्रत्यय	तिप् (tip)	गम् + तिप्
4	Apply phonetic rules	गुण, वृद्धि, संधि, आदि	गच्छति (gacchati)

LIKE A PROGRAM EXECUTION

```
root = "gam";
tense = "lat"; // present
affix = "tip"; // 3rd person singular
word = applyRules(root, tense, affix);
output = "gacchati";
```

WHY IT MATTERS TODAY

- Inspires Natural Language Processing and AI.
- Basis for rule-based systems and formal grammars.
- Shows the power of human intellect in creating algorithms.
- A bridge between ancient wisdom and modern computation.

IN A NUTSHELL

Aṣṭādhyāyī is a genius computational system that:

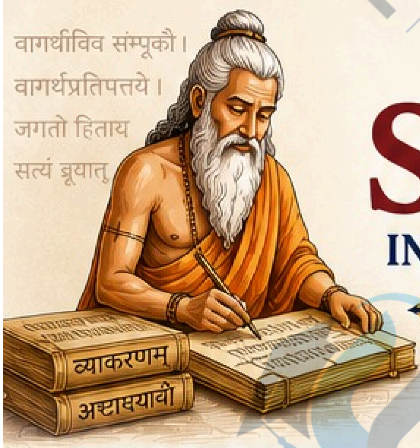
- ✔ Accepts input (roots)
- ✔ Applies rules (operations)
- ✔ Generates output (words/sentences)
- ✔ Ensures correctness (constraints)
- ✔ Works efficiently (optimization)

"Pāṇini was not just a grammarian, he was the world's first programmer.

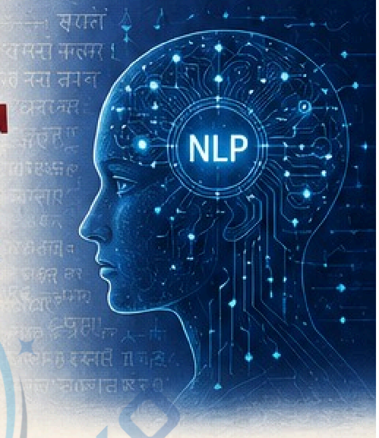
– A timeless contribution to computational thinking."

यथा पाणिनिना भाषायाः सूत्ररूपेण व्यवस्था कृता ।
तथा अद्य तस्माद् विज्ञानं कम्प्यूटरशास्त्रं प्रेरणा लभते ॥

As Pāṇini structured language in the form of rules, today computer science gets inspiration from it.



ROLE OF SANSKRIT IN NATURAL LANGUAGE PROCESSING



Ancient Wisdom. Modern Intelligence.

Sanskrit – a highly structured, precise and logical language – provides a strong foundation for the science and technology of language.

WHAT IS NLP?

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret and generate human language.

Human Language → NLP System → Understanding & Meaning

WHY SANSKRIT MATTERS FOR NLP?

- Highly structured and rule-based
- Unambiguous and precise
- Rich morphological system
- Well-defined grammar by Pāṇini
- Phonetic consistency (sound-to-script mapping)
- Large knowledge base in literature and science

KEY FEATURES OF SANSKRIT THAT BENEFIT NLP

1. FORMAL GRAMMAR

Pāṇini's Aṣṭādhyāyī is a rule-based generative grammar – like an algorithm.

Example: Rules generate infinite correct sentences.

2. MORPHOLOGICAL RICHNESS

Words are formed using roots, prefixes, suffixes and inflections in a systematic way.

Example:
गम् (gam – go)
+ति (ti – 3rd person sing.)
= गच्छति (gacchati – goes)

3. COMPOUNDING (SAMĀSA)

Sanskrit allows creation of precise compound words that capture complex meanings.

Example: राजपुरुषः (rāja-puruṣaḥ) = King's minister

4. FREE WORD ORDER

Meaning is preserved due to rich inflectional endings, enabling flexible word order.

Example:
रामः वनं गच्छति
रामं वनं गच्छति
वनं रामः गच्छति
(All mean: Rama goes to the forest)

5. PHONETIC PRECISION

One sound = one symbol. One symbol = one sound. No ambiguity in pronunciation.

Example:
"क" is always /ka/
"ग" is always /ga/

6. SEMANTIC CLARITY

Words have well-defined meanings with relations between concepts clearly expressed.

Example: धर्मः (dharma) – duty, righteousness

APPLICATIONS OF SANSKRIT IN NLP

1. MACHINE TRANSLATION

Sanskrit's clarity and structure improve accuracy in translation systems.

2. MORPHOLOGICAL ANALYSIS

Rule-based morphological analyzers are highly effective for Sanskrit word forms.

3. PART-OF-SPEECH TAGGING

NOUN VERB ADJ

Well-defined word classes and endings help in accurate POS tagging.

4. PARSING & SYNTAX ANALYSIS

Sanskrit grammar provides clear rules for building parse trees and dependencies.

5. INFORMATION EXTRACTION

Structured texts (e.g., śāstras, purāṇas) can be mined for knowledge.

6. TEXT SUMMARIZATION

Compound structures and semantic clarity help in better abstractive summarization.

7. SPEECH TECHNOLOGY

Phonetic consistency helps in accurate ASR (Automatic Speech Recognition) and TTS systems.

8. ONTOLOGY & KNOWLEDGE GRAPHS

Sanskrit's rich conceptual system supports building ontologies and knowledge graphs.

EXAMPLE: SANSKRIT STRUCTURE IN NLP

Sanskrit Expression	Analysis	NLP Benefit
गच्छति (gacchati)	gam (root) + laṭ (tense: present) + ti (3rd person singular)	Easy morphological parsing and generation
बालकः पठति	bālakaḥ (boy) + paṭhati (reads)	Clear subject-verb relation detection
राजपुरुषः आगच्छति	rāja-puruṣaḥ (king's minister) + āgacchati (arrives)	Compound word understanding
सर्वे जनाः सुखिनो भवन्तु	sarve janāḥ sukhino bhavantu (May all people be happy)	Sentiment & intent recognition

BENEFITS OF USING SANSKRIT IN NLP

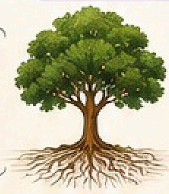
- Reduces ambiguity in language understanding
- Improves accuracy of parsing and generation
- Enhances low-resource language technologies
- Preserves India's linguistic heritage in AI
- Bridges ancient knowledge with cutting-edge technology

संस्कृतम् विश्वस्य भवितव्यस्य भाषा।
– महर्षि पाणिनि

Sanskrit is a language for the future of the world.
– Maharshi Panini

THE FUTURE IS ROOTED IN OUR HERITAGE

Sanskrit is not just a classical language; it is a computationally perfect language. By integrating Sanskrit with NLP, we can build smarter, more reliable and culturally rooted AI systems.



WHERE ANCIENT MEETS INTELLIGENCE

Language Technology · AI & Machine Learning · Cultural Preservation · Knowledge Discovery · Human-Centric AI

॥ यत्र विश्वं भवत्येकनीडम् ॥
"Where the world becomes one nest."

IKS IN SCIENCE, ENGINEERING AND TECHNOLOGY

Ancient Wisdom • Timeless Principles • Modern Innovations

विद्या ददाति विनयम्

सा विद्या या विमुक्तये

That knowledge which liberates and leads to well-being for all.

The Indian Knowledge System (IKS) integrates observation, experimentation, logic and ethics for the welfare of all beings and the environment.

1. IKS IN SCIENCE

IKS provides a holistic understanding of the natural world through systematic observation, experimentation and empirical reasoning.

Astronomy (Jyotiṣa)
Accurate calculations of planetary motions, eclipses and calendars.
Examples: Āryabhaṭīya, Sūrya Siddhānta

Mathematics (Gaṇita)
Concepts of zero, decimals, place value, algebra, permutations, combinations and infinite series.
Examples: Baudhāyana, Bhāskara II

Physics (Bhautika)
Insights into matter, energy, sound, light and motion.
Examples: Vaiśeṣika Sūtra, Nyāya Sūtra

Chemistry (Rasaśāstra)
Metallurgy, alchemy, herbal chemistry, nano-particles and sustainable processes.
Examples: Rasa Ratna Samuccaya

Biology & Life Sciences (Jīva Vigyāna)
Ayurveda, plant sciences, ecology and understanding of life processes.
Examples: Caraka Saṃhitā, Suśruta Saṃhitā

2. IKS IN ENGINEERING

IKS texts describe advanced engineering principles, sustainable design and large-scale infrastructure for societal needs.

Architecture (Vāstu & Śilpa)
Scientific principles of planning, orientation, load management, aesthetics and harmony with nature.
Examples: Mānasāra, Mayamata

Civil Engineering
Dams, step wells, canals, roads and urban planning with environmental balance.
Examples: Bāori, Irrigation systems, Rāja-mārga

Mechanical Engineering
Machines, mechanisms, robotics and automated systems.
Examples: Yantra-s, Vaimānika Śāstra

Aerospace Engineering
Concepts of flight, aerodynamics, aircraft design and propulsion.
Examples: Vaimānika Śāstra

Materials & Metallurgy
High-quality steels (Wootz), corrosion resistant alloys and metal treatment.
Examples: Zinc distillation, Iron pillar of Delhi

3. IKS IN TECHNOLOGY

IKS inspires innovative technologies that are sustainable, human-centric and aligned with ethical responsibility.

Information Technology
Algorithms, compression, data structures and knowledge representation.
Examples: Pingala's binary, Panini's grammar as a generative system

Communication Technology
Acoustics, signal systems, encryption and wireless ideas.
Examples: Nāda (sound) science, Gūḥya sūtras (coded messages)

Energy Technology
Solar, wind, biogas and water energy systems for sustainable living.
Examples: Sūrya upāsanā, Biomass usage, Water wheels

Medical Technology
Surgical instruments, prosthetics, herbal formulations and diagnostics.
Examples: Suśruta's instruments, Ayurvedic formulations

Environmental Technology
Waste management, water purification, afforestation and biodiversity conservation.
Examples: Sacred groves, Traditional water purification

CORE PRINCIPLES OF IKS

Holistic & Integrated — Harmony of body, mind, society and nature

Empirical & Experiential — Observation, experimentation and validation

Logical & Analytical — Strong reasoning, systematic thinking and modeling

Ethical & Sustainable — Knowledge for the welfare of all and future generations

Timeless & Universal — Applicable across time, place and disciplines

People-centric & Value-based — Focus on dharma, well-being and social harmony

EXAMPLES OF IKS TO MODERN APPLICATIONS

IKS Contribution	Modern Relevance
Zero (Śūnya)	Computing, Cryptography, Digital Systems
Decimal System & Place Value	All modern mathematics and computing
Pingala's Binary (Chandas Śāstra)	Binary code in computers, digital logic
Vāstu Principles	Green buildings, sustainable architecture
Ayurveda & Rasaśāstra	Pharmaceutical research, nanomedicine
Ancient Water Systems	Sustainable water management practices
Panini's Grammar	Natural Language Processing, AI, Linguistics
Dhātu Science & Metallurgy	Advanced materials, corrosion resistance

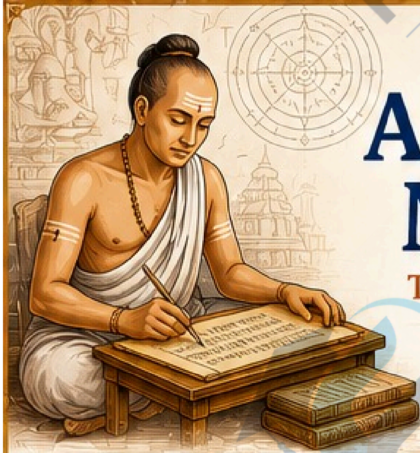
WHY IKS MATTERS TODAY

- Provides sustainable and eco-friendly solutions
- Encourages innovation with indigenous thinking
- Strengthens self-reliance in science and technology
- Supports ethical AI and human-centric technology
- Preserves cultural heritage while solving modern challenges
- Builds a harmonious and resilient future

लोकाः समस्ताः सुखिनो भवन्तु
May all beings be happy and free.

IKS is not just our past, it is the foundation for a better and wiser future.

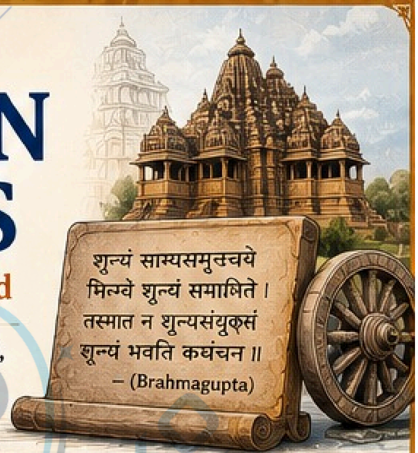
When ancient wisdom meets modern science, innovation becomes meaningful and progress becomes sustainable.



INTRODUCTION TO ANCIENT INDIAN MATHEMATICS

The Timeless Legacy of Numbers and Beyond

Ancient Indian mathematics is a rich tradition of logical thinking, abstract reasoning and practical application that shaped the world's mathematical knowledge.



HISTORICAL OVERVIEW

- Indian mathematics has a history of more than 2000 years.
- It developed not only for theoretical curiosity but also for practical needs in astronomy, engineering, trade, architecture and daily life.
- Great mathematicians like Āryabhaṭa, Brahmagupta, Bhāskara II and others made priceless contributions.

KEY FEATURES

- **Place Value System** – The systematic use of place value.
- **Concept of Zero (Śūnya)** – India gave the world the idea of zero.
- **Positive & Negative Numbers** – Early understanding and rules.
- **Advanced Algebra** – Solving linear, quadratic equations and indeterminate equations.
- **Trigonometry** – Accurate values of trigonometric functions.

PROMINENT MATHEMATICIANS

Āryabhaṭa (476–550 CE)
Father of Indian mathematics and astronomy. He calculated the value of π and explained place value system.

Brahmagupta (598–668 CE)
Gave rules for operations with zero and negative numbers. Wrote *Brahmasphutasiddhānta*.

Bhāskara II (1114–1185 CE)
Author of *Siddhānta Śiromaṇi*. Made important work in algebra, calculus concepts and astronomy.

Varāhamihira, Lalla, Mahāvira, Nīlakaṇṭha and others also made remarkable contributions.

MAJOR CONTRIBUTIONS OF ANCIENT INDIAN MATHEMATICS

1. THE NUMBER SYSTEM

Development of the decimal place value system.

1	2	3	4
10	20	30	40
100	200	300	400
1000	2000	3000	4000

Use of 0 as a placeholder changed the world.

2. ZERO (ŚŪNYA)

India gave the world the concept of zero.

0

"Śūnya is the root of all number systems."
– Brahmagupta

3. ARITHMETIC

Rules for arithmetic operations including with zero and negative numbers.

$$a + 0 = a$$
$$a - 0 = a$$
$$a \times 0 = 0$$
$$a + (-b) = a - b$$
$$a - (-b) = a + b$$

4. ALGEBRA (BĪJA GANITA)

Solving linear and quadratic equations.

Example:

Solve $x^2 + 10x = 39$

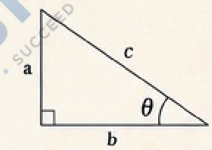
$$x^2 + 10x - 39 = 0$$
$$(x + 13)(x - 3) = 0$$
$$x = -13 \quad \text{or} \quad x = 3$$

5. TRIGONOMETRY

Accurate values and identities.

Example:

$$\sin^2\theta + \cos^2\theta = 1$$



$\sin\theta = \frac{a}{c}$, $\cos\theta = \frac{b}{c}$, $\tan\theta = \frac{a}{b}$

APPLICATIONS IN DAILY LIFE AND SCIENCE

- **Astronomy** – Calculating planetary positions, eclipses and calendars.
- **Architecture** – Measurement, design and construction of temples, cities and structures.
- **Trade & Commerce** – Calculations in weights, measures, interest and exchange.
- **Engineering** – Water management, irrigation systems, bridges and mechanical devices.

NOTABLE WORKS

- **Āryabhaṭiya** – By Āryabhaṭa (Astronomy and Mathematics)
- **Brahmasphutasiddhānta** – By Brahmagupta (Mathematics and Astronomy)
- **Siddhānta Śiromaṇi** – By Bhāskara II (Mathematics, Algebra, Calculus)
- **Lilāvati** – By Bhāskara II (Arithmetic and Algebra)

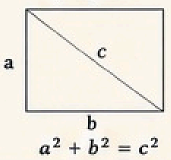
UNIQUE ASPECTS

- **Holistic approach** – Mathematics connected with astronomy, philosophy and nature.
- Highly systematic and logical methods.
- Emphasis on problem solving and generalization.
- Development of infinite series and early ideas of calculus.
- **Universal impact** – Influenced Arabic, Persian and European mathematics.

BEAUTIFUL EXAMPLES FROM ANCIENT INDIAN MATHEMATICS

1. SULBA SUTRAS
(Geometry)

Pythagoras theorem was known to Indians long before Pythagoras.



$$a^2 + b^2 = c^2$$

2. INFINITE SERIES
(Āryabhaṭa)

Sum of natural numbers:

$$1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$$

Sum of squares:

$$1^2 + 2^2 + 3^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{}$$

3. APPROXIMATION OF π
(Āryabhaṭa)

$$\pi \approx 3.1416$$

(Correct up to 4 decimal places)

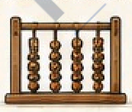
π

4. KUTTĀKA METHOD
(Brahmagupta)

A method to solve indeterminate equations of the form:

$$ax + by = c$$

Used in algebra and number theory.



65. BINOMIAL COEFFICIENTS
(Meru Prastāra)

Early study of combinations.

Pascal's triangle was known as Meru Prastāra.

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1

OUR HERITAGE, OUR PRIDE

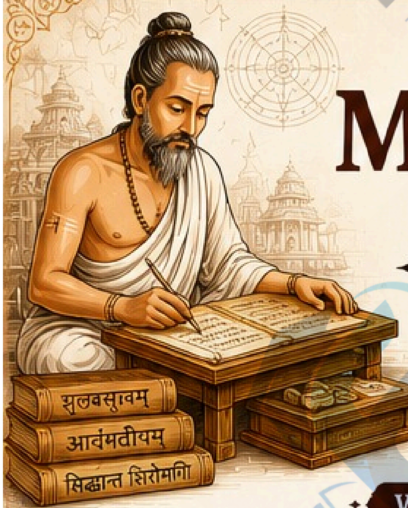
Ancient Indian mathematics is not just our past, it is a guiding light for the future of mathematics, science, technology and innovation.

"The genius of Indian mathematicians continues to inspire the world.
Let us preserve, learn and innovate."

LEGACY FOR THE FUTURE

From zero to infinity, from numbers to the stars – Indian mathematics continues to empower humanity's progress.

IMPORTANCE OF MATHEMATICS IN ANCIENT TIME



THE LANGUAGE OF LOGIC THAT BUILT A CIVILIZATION

In ancient India, mathematics was not just a subject, it was a way of life. It helped in understanding the universe, solving real-life problems and building a prosperous society.

गणितं ज्ञानम् न केवलं
पण्डितानां कृते,
अपि तु सर्वेभ्यः
जीवनोपयोगि च ॥

– Mathematics is not only for scholars, but is useful in everyone's life.

WHY MATHEMATICS WAS IMPORTANT IN ANCIENT TIME?

1. UNDERSTANDING THE UNIVERSE
Mathematics helped ancient scholars study the movement of planets, stars, eclipses and seasons with great accuracy.

2. DAILY LIFE APPLICATIONS
Used in counting, measuring, trading, farming, cooking, division of land and sharing of resources.

3. ARCHITECTURE AND ENGINEERING
Essential in designing and constructing temples, palaces, stepwells, forts, roads, bridges and irrigation systems.

4. TRADE AND COMMERCE
Mathematics enabled fair trade, calculation of profit, interest, weights, measures and currency exchange.

5. SCIENCE AND TECHNOLOGY
Foundation for astronomy, medicine, metallurgy, navigation, calendar making and many scientific fields.

6. SPIRITUAL AND PHILOSOPHICAL VALUE
ॐ

Mathematics was seen as a path to truth, order and harmony in the universe.

MATHEMATICS IN ACTION – REAL LIFE EXAMPLES

CALENDARS
Used for Panchang, festivals, eclipses and agricultural planning.

LAND MEASUREMENT (SULBA SUTRAS)
Accurate measurement of land, construction of altars and buildings.

IRRIGATION SYSTEMS
Designing canals, reservoirs and stepwells using geometry and calculation.

ACCOUNTING & RECORD KEEPING
Use of place value system, fractions and arithmetic in accounts and records.

NAVIGATION
Mathematics helped in finding directions, distance and time while traveling.

GAMES & PUZZLES
Development of logic, patterns and strategies through games.

GREAT INDIAN MATHEMATICIANS

ĀRYABHATA (476–550 CE)
Introduced zero, place value system, trigonometric tables and astronomy.

BRAHMAGUPTA (598–668 CE)
Gave rules for zero and negative numbers, solutions of equations, area of cyclic quadrilateral.

BHĀSKARA II (1114–1185 CE)
Work on algebra, calculus concepts, sine approximation and astronomical calculations.

VARĀHAMIHIRA (505–587 CE)
Used mathematics in astronomy, calendar and weather prediction.

MAJOR MATHEMATICAL ACHIEVEMENTS

123 — **PLACE VALUE SYSTEM**
Revolutionary system that forms the base of modern mathematics.

0 — **THE CONCEPT OF ZERO (ŚŪNYA)**
A unique gift to the world that changed mathematics forever.

$\frac{1}{2}$ — **FRACTIONS AND DECIMALS**
Use of proper and improper fractions, and early ideas of decimals.

x^2 — **ALGEBRA (BĪJA GANITA)**
Solving equations, indeterminate equations and quadratic problems.

$\sin\theta$ — **TRIGONOMETRY**
Development of sine, cosine, versine tables and trigonometric identities.

IMPACT ON THE WORLD

- Indian mathematical ideas spread to the Arab world and Europe.
- Influenced modern mathematics, science and technology.
- Built the foundation for innovation, discovery and progress.
- Improved quality of life and societal development.

IN SUMMARY

- Mathematics was the backbone of ancient Indian knowledge and progress.
- It connected science, society, spirituality and daily life.
- It reflects the wisdom, creativity and practical thinking of our ancestors.

Mathematics was not just calculated... It was celebrated!

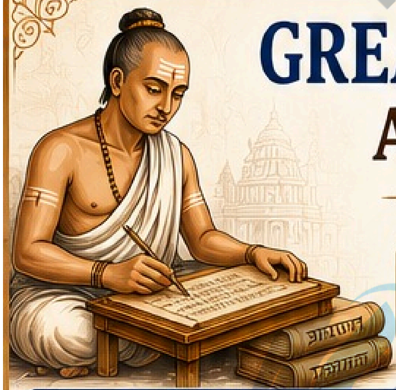
A TIMELESS LEGACY

From counting grains to calculating the cosmos, from building temples to exploring infinity – mathematics was the key to knowledge and the engine of civilization.

Our heritage. Our pride.

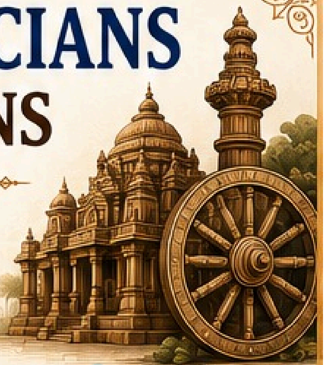
ANCIENT WISDOM, ETERNAL RELEVANCE

The numbers may change, but the importance of mathematics remains forever.

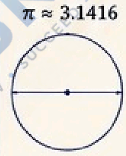
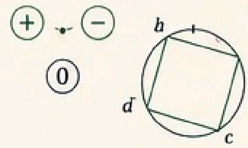
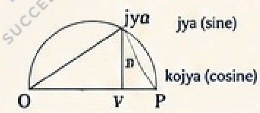
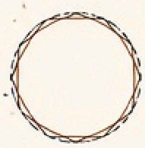
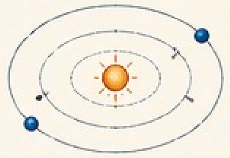
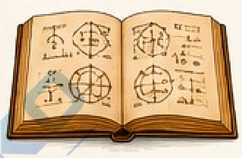


GREAT INDIAN MATHEMATICIANS AND THEIR CONTRIBUTIONS

Brilliant Minds • Timeless Ideas • Global Impact



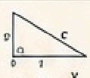
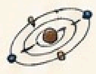
Ancient India produced some of the world's greatest mathematicians whose ideas laid the foundation of modern mathematics and continue to inspire the world.

MATHEMATICIAN	PERIOD	MAJOR CONTRIBUTIONS	KEY ACHIEVEMENTS / EXAMPLES
1  **ĀRYABHAṬA** (476 – 550 CE) Author of *Āryabhaṭīya*	5th – 6th Century	• Gave the value of π (pi) approximately. • Introduced place value system. • Explained zero as a number. • Worked on algebra, trigonometry and astronomical calculations.	**Key Achievements / Examples** • Value of $\pi \approx 3.1416$ • Place value: Units, Tens, Hundreds... • Solved linear equations. $\pi \approx 3.1416$  $1002 = 1\times10^3 + 0\times10^2 + 0\times10^1 + 2\times10^0$
2  **BRAHMAGUPTA** (598 – 668 CE) Author of *Brahmasphuṭasiddhānta*	7th Century	• Systematically described arithmetic operations with zero. • Gave rules for positive and negative numbers. • Solved quadratic equations. • Worked on geometry and cyclic quadrilaterals.	**Key Achievements / Examples** • Rules for operations with zero. • Solution of $ax^2 + bx = c$. • Area of cyclic quadrilateral: $\sqrt{(s-a)(s-b)(s-c)(s-d)}$ (Brahmagupta's Formula) 
3  **BHĀSKARA I** (7th Century) Author of *Mahābhāskarīya*	7th Century	• Explained arithmetic, algebra, and mensuration. • Worked on indeterminate equations (kuttaka method). • Calculated sine tables.	**Key Achievements / Examples** • Kuttaka method for solving $ax + by = c$. • Improved value of π. $ax + by = c$ (Kuttaka Method) θ: 0°, 30°, 45°, 60°, 90° sin θ: 0, $\frac{1}{2}$, $\frac{1}{\sqrt{2}}$, $\frac{\sqrt{3}}{2}$, 1
4  **BHĀSKARA II** (1114 – 1185 CE) Author of *Siddhānta Śiromaṇi*	12th Century	• Made great advances in algebra, calculus and trigonometry. • Gave the concept of derivatives (differential calculus). • Solved complex equations.	**Key Achievements / Examples** • Concept of derivative: $\frac{d(y)}{dx}$ • Chakravāla method for Pell's equation. **Pell's Equation** $x^2 - Dy^2 = 1$ 
5  **MĀDHAVA OF SAṄGAMAGRĀMA** (1340 – 1425 CE) Founder of the Kerala School	14th – 15th Century	• Developed infinite series for π, sine, cosine, arctangent. • Worked on approximations with great accuracy. • Laid foundation for calculus.	**Key Achievements / Examples** • Series for π: $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots$ • Infinite series expansions. 
6  **NĪLAKAṆṬHA SOMAYĀJĪ** (1444 – 1544 CE) Author of *Tantra Saṅgraha*	15th – 16th Century	• Refined astronomical models. • Calculated accurate value of π. • Worked on infinite series and planetary motions.	**Key Achievements / Examples** • Value of π correct to 7 decimals: $\pi \approx 3.1415926$ • Improved sine tables. 
7  **JAGADĪŚA & NĀRĀYAṆA PANDITA** (16th Century) Authors of *Gaṇita Kaumudī*	16th Century	• Wrote an important text on mathematics. • Covered arithmetic, algebra, geometry, mensuration and astronomy. • Helped preserve and teach Indian mathematical knowledge.	**Key Achievements / Examples** • Comprehensive mathematics textbook. • Simplified complex topics for learners. 

LEGACY

- Indian mathematicians introduced concepts that were centuries ahead of their time.
- Their work influenced Arab scholars and later reached Europe.
- Their ideas continue to inspire modern mathematics, science and technology.

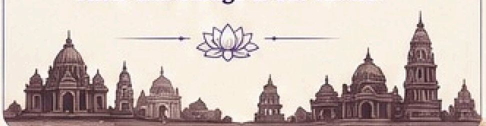
CONTRIBUTIONS THAT CHANGED THE WORLD

0 Zero	 Place Value System	$ax^2 + bx + c = 0$ Algebra	 $\sin\theta = \frac{y}{r}$ $\cos\theta = \frac{x}{r}$
$\sum_{n=1}^{\infty} a_n$ Infinite Series	$\frac{d(y)}{dx}$ Calculus (early ideas)	 Astronomy & Planetary Models	π Approximations of π

A TIMELESS INSPIRATION

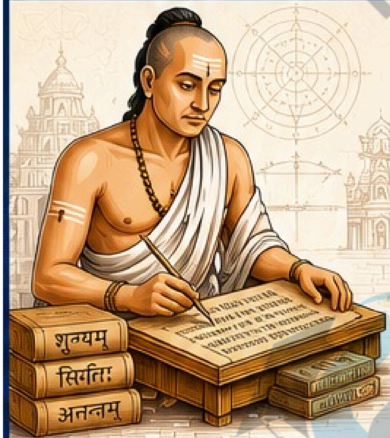
These brilliant minds from ancient India proved that when curiosity meets logic, the possibilities are infinite.

Their wisdom is our pride.
Their knowledge is our future.



"From ancient wisdom to modern innovation – Indian mathematics lights the path of knowledge."

NUMBER SYSTEM
AND ITS SIGNIFICANCE



THE LANGUAGE OF MATHEMATICS & THE FOUNDATION OF ALL COUNTING

A Number System is a way to represent numbers using a set of digits or symbols according to specific rules.
It is the base of all mathematical calculations and real-life applications.

शून्यं समारस्पते सर्वः
तस्माद शून्यं प्रसस्यते ।
शून्याद अन्यतुं न किव्चन्
शून्याय विश्वं प्रतिष्ठितम् ॥
– (Brahmagupta)

TYPES OF NUMBER SYSTEMS

1. NATURAL NUMBERS
Counting numbers starting from 1.
1, 2, 3, 4, 5, ...
Used in counting objects.

2. WHOLE NUMBERS
Natural numbers including zero.
0, 1, 2, 3, 4, 5, ...
Used in counting and numbering.

3. INTEGERS
Whole numbers and their negatives.
..., −3, −2, −1, 0, 1, 2, 3, ...
−3 −2 −1 0 1 2 3
Used in real-life situations like loss, profit, temperature, etc.

4. RATIONAL NUMBERS
Numbers in the form p/q where $p, q \in \mathbb{Z}$, $q \neq 0$
$\frac{1}{2}, -\frac{3}{4}, \frac{5}{7}, \ldots$
Used in fractions, decimals (terminating or repeating).

5. IRRATIONAL NUMBERS
Numbers that cannot be expressed in p/q form.
$\sqrt{2}, \sqrt{3}, \pi, e, \ldots$
Used in geometry, advanced mathematics and real measurements.

6. REAL NUMBERS
All rational and irrational numbers.
$R = Q \cup (I - Q)$
R Q I
Covers all numbers on the number line.

$-\infty$ −4 −2 −1 0 1 2 3 4 $+\infty$

SIGNIFICANCE OF NUMBER SYSTEM

1. FOUNDATION OF MATHEMATICS
All mathematical operations, formulas and theories are built upon number systems.

2. USED IN EVERYDAY LIFE
Counting, money, measurement, time, age, distance – everything is based on numbers.

3. SCIENCE & ENGINEERING
Essential in physics, chemistry, computer science, engineering, and data analysis.

4. TECHNOLOGY & COMPUTING
10101
Binary number system is the base of computers, digital devices and modern electronics.

5. FINANCE & ECONOMICS
Used in banking, accounting, trading, statistics and economic modeling.

6. MEASUREMENT & NAVIGATION
Used in surveying, GPS, space science, astronomy and mapping.

Zero (0) – The greatest gift of ancient India to the world. It made complex calculations possible and led to the development of modern mathematics, science and technology.

KEY FEATURES OF NUMBER SYSTEMS

- 1 2 / 2 3 — Provide a systematic way to represent and organize numbers.
- Make calculations easier, faster and more accurate.
- Help in solving real-life problems and making decisions.
- Allow us to work with very large or very small numbers.

PLACE VALUE SYSTEM – OUR UNIQUE CONTRIBUTION

The place value system with 10 digits (0–9) allows us to represent any number using fewer symbols.

Thousands	Hundreds	Tens	Ones		Tenths	Hundredths
1	2	3	4	.	4	5
1000	200	30	4		0.5	0.05

Example:
1234.45
means
$1 \times 1000 + 2 \times 100 + 3 \times 10 + 4 \times 1 + 4 \times 0.1 + 5 \times 0.01$

This system was first clearly explained in ancient Indian texts by mathematicians like Brahmagupta, Bhāskara II and others.

INTERESTING FACTS

- 0 — The concept of Zero was first discovered in India.
- 10 — The decimal (base-10) number system originated in India.
- ∞ — Indian mathematicians worked with very large numbers and infinite series.
- Our number system spread to the Arab world and later to Europe.

IMPORTANCE IN MODERN WORLD

Number System – The Invisible Power Behind Everything

- Smartphones
- Artificial Intelligence
- Internet & Data
- Robotics
- Space Exploration
- Medical Technology

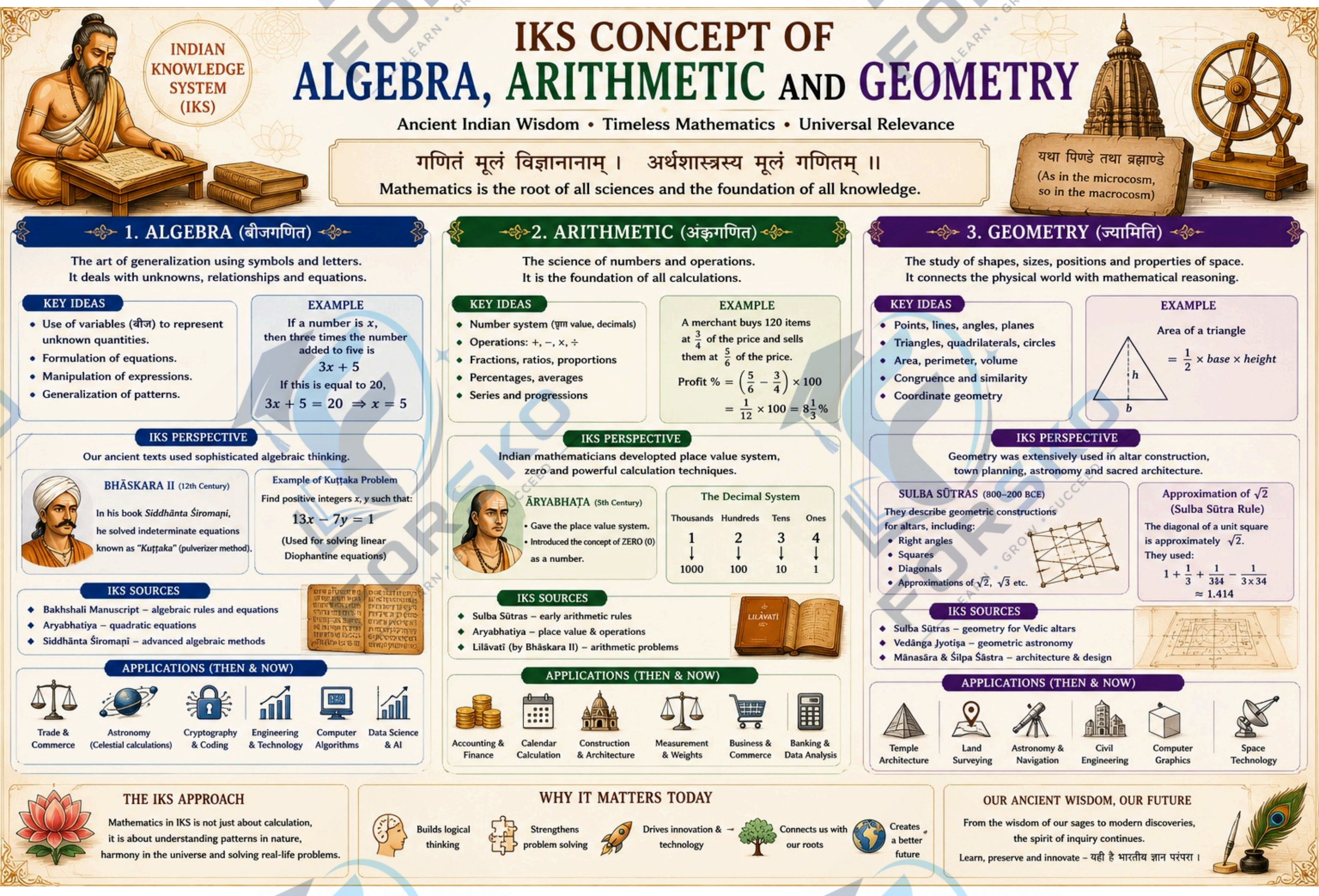
CONCLUSION

Number system is not just about numbers, it is the cornerstone of human progress.

From ancient calculations to modern technology, it continues to shape our world and future.

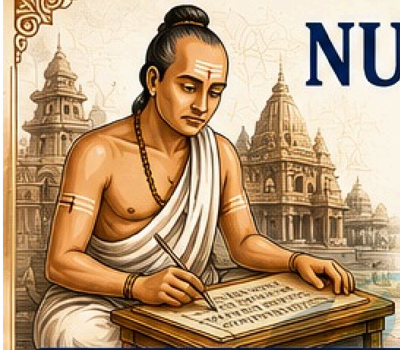
Understand numbers, Understand the world.

"Where there are numbers, there is order; where there is order, there is progress." – Āryabhaṭa


INDIAN KNOWLEDGE SYSTEM (IKS)
IKS CONCEPT OF
ALGEBRA, ARITHMETIC AND GEOMETRY
Ancient Indian Wisdom • Timeless Mathematics • Universal Relevance
गणितं मूलं विज्ञानानाम् । अर्थशास्त्रस्य मूलं गणितम् ॥
Mathematics is the root of all sciences and the foundation of all knowledge.
यथा पिण्डे तथा ब्रह्माण्डे
(As in the microcosm, so in the macrocosm)
1. ALGEBRA (बीजगणित)
The art of generalization using symbols and letters.
It deals with unknowns, relationships and equations.
KEY IDEAS
• Use of variables (बीज) to represent unknown quantities.
• Formulation of equations.
• Manipulation of expressions.
• Generalization of patterns.
EXAMPLE
If a number is x, then three times the number added to five is
$3x + 5$
If this is equal to 20,
$3x + 5 = 20 \Rightarrow x = 5$
IKS PERSPECTIVE
Our ancient texts used sophisticated algebraic thinking.
BHĀSKARA II (12th Century)
In his book *Siddhānta Śiromaṇi*, he solved indeterminate equations known as "*Kuṭṭaka*" (pulverizer method).
Example of Kuṭṭaka Problem
Find positive integers x, y such that:
$13x - 7y = 1$
(Used for solving linear Diophantine equations)
IKS SOURCES
◆ Bakhshali Manuscript – algebraic rules and equations
◆ Aryabhatiya – quadratic equations
◆ Siddhānta Śiromaṇi – advanced algebraic methods
APPLICATIONS (THEN & NOW)
Trade & Commerce
Astronomy (Celestial calculations)
Cryptography & Coding
Engineering & Technology
Computer Algorithms
Data Science & AI
2. ARITHMETIC (अंङ्कगणित)
The science of numbers and operations.
It is the foundation of all calculations.
KEY IDEAS
◆ Number system (स्थान value, decimals)
◆ Operations: +, −, ×, ÷
◆ Fractions, ratios, proportions
◆ Percentages, averages
◆ Series and progressions
EXAMPLE
A merchant buys 120 items at $\frac{3}{4}$ of the price and sells them at $\frac{5}{6}$ of the price.
Profit % $= \left(\frac{5}{6} - \frac{3}{4}\right) \times 100$
$= \frac{1}{12} \times 100 = 8\frac{1}{3}\%$
IKS PERSPECTIVE
Indian mathematicians developed place value system, zero and powerful calculation techniques.
ĀRYABHAṬA (5th Century)
• Gave the place value system.
• Introduced the concept of ZERO (0) as a number.
The Decimal System
Thousands Hundreds Tens Ones
1 2 3 4
1000 100 10 1
IKS SOURCES
◆ Sulba Sūtras – early arithmetic rules
◆ Aryabhatiya – place value & operations
◆ Lilāvatī (by Bhāskara II) – arithmetic problems
LILĀVATĪ
APPLICATIONS (THEN & NOW)
Accounting & Finance
Calendar Calculation
Construction & Architecture
Measurement & Weights
Business & Commerce
Banking & Data Analysis
3. GEOMETRY (ज्यामिति)
The study of shapes, sizes, positions and properties of space.
It connects the physical world with mathematical reasoning.
KEY IDEAS
• Points, lines, angles, planes
• Triangles, quadrilaterals, circles
• Area, perimeter, volume
• Congruence and similarity
• Coordinate geometry
EXAMPLE
Area of a triangle
$= \frac{1}{2} \times base \times height$
h
b
IKS PERSPECTIVE
Geometry was extensively used in altar construction, town planning, astronomy and sacred architecture.
SULBA SŪTRAS (800–200 BCE)
They describe geometric constructions for altars, including:
• Right angles
• Squares
• Diagonals
• Approximations of $\sqrt{2}$, $\sqrt{3}$ etc.
Approximation of $\sqrt{2}$ (Sulba Sūtra Rule)
The diagonal of a unit square is approximately $\sqrt{2}$.
They used:
$1 + \frac{1}{3} + \frac{1}{3\cdot4} - \frac{1}{3\times34}$
≈ 1.414
IKS SOURCES
◆ Sulba Sūtras – geometry for Vedic altars
◆ Vedānga Jyotiṣa – geometric astronomy
◆ Mānasāra & Śilpa Śāstra – architecture & design
APPLICATIONS (THEN & NOW)
Temple Architecture
Land Surveying
Astronomy & Navigation
Civil Engineering
Computer Graphics
Space Technology
THE IKS APPROACH
Mathematics in IKS is not just about calculation, it is about understanding patterns in nature, harmony in the universe and solving real-life problems.
WHY IT MATTERS TODAY
Builds logical thinking
Strengthens problem solving
Drives innovation & technology
Connects us with our roots
Creates a better future
OUR ANCIENT WISDOM, OUR FUTURE
From the wisdom of our sages to modern discoveries, the spirit of inquiry continues.
Learn, preserve and innovate – यही है भारतीय ज्ञान परंपरा ।

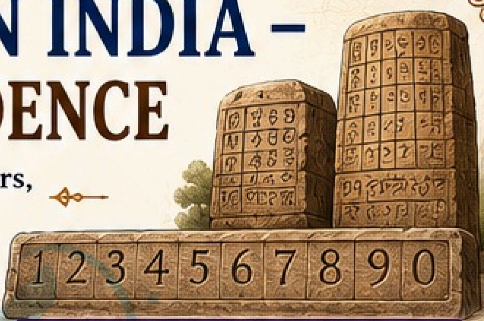
Unit - III


DATA STRUCTURES
ALGORITHMS
COMPUTER NETWORKS
DATABASE SYSTEMS
#include <iostream>
using namespace std;
int main()
{
cout << "Hello, World!";
return 0;
}



NUMBER SYSTEM IN INDIA – HISTORICAL EVIDENCE

India gave the world not just numbers, but a complete system of thought.



INDIA – THE BIRTHPLACE OF OUR NUMBERS

- The decimal place value system and the symbol for ZERO (0) were developed in India.
- Indian mathematicians created a logical, positional system that made calculations simple and powerful.
- This innovation changed mathematics forever and became the foundation of modern arithmetic, algebra and computer science.

EVOLUTION OF INDIAN NUMERALS

From Symbols to Our Modern Digits

Brahmi (3rd century BCE)	Ƭ	⁙	<	।	C	Ł	ο	Ɛ	ο
Gupta (4th–6th century CE)	ŧ	2	3	५	ℶ	Z	6	8	٥
Siddham (7th–9th century CE)	ŧ	2	3	4	5	5	2	8	6
Devanagari (10th century CE)	1	ᄅ	3	4	5	७	7	8	९
Modern Digits (Used Worldwide)	1	2	3	4	5	6	7	8	9 0

THE SPECIAL GIFT OF INDIA – ZERO

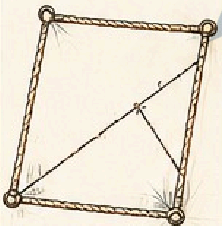
- Zero is not just a number, it is a concept.
- First clear use of ZERO as a number and as a placeholder in positional notation is found in India.
- It made it possible to represent large numbers and perform complex calculations.
- The world adopted zero through Indian–Arab transmission.

0
शून्य
(ZERO)

HISTORICAL EVIDENCE FROM ANCIENT INDIAN SOURCES

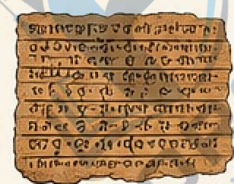
1. SULBA SUTRAS (800–200 BCE)

Early geometrical texts used for altar construction. Evidence of understanding of integers, fractions, square roots and Pythagorean triples.



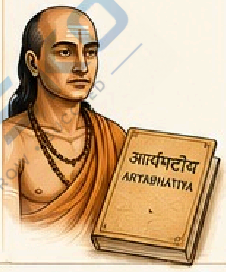
2. BAKSHALI MANUSCRIPT (3rd – 7th century CE)

One of the earliest manuscripts on mathematics. Contains arithmetic operations, negative numbers, zero and solutions of quadratic equations.



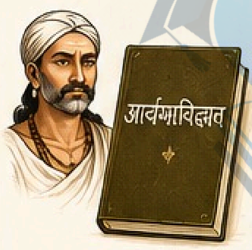
3. ARYABHATIYA (499 CE)

By Aryabhata. Explains place value system, zero, arithmetic operations, and large numbers using the decimal system.



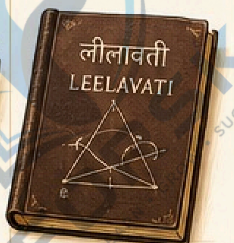
4. BRAHMASPHUTASIDDHANTA (628 CE)

By Brahmagupta. Clear rules for arithmetic with zero and negative numbers. Advanced methods for solving equations.



5. LEELAVATI (12th century CE)

By Bhaskaracharya II. Rich in arithmetic, algebra, geometry and practical problems with systematic methods.



6. INSCRIPTIONS & ARCHAEOLOGICAL EVIDENCE

Many temple inscriptions, coins and ancient records contain numbers written in early Indian numeral forms, proving their widespread use.

KEY FEATURES OF THE INDIAN NUMBER SYSTEM

- 10 **Decimal System (Base 10)** Uses 10 digits: 0–9
- **Place Value System** The value of a digit depends on its position.
- 0 **Zero as a Number & Placeholder** A revolutionary concept unique to India.
- 10^n **Supports Large Numbers** Allows easy representation and calculation of very large and very small numbers.
- **Algorithmic Operations** Simple rules for arithmetic, algebra and beyond.

JOURNEY FROM INDIA TO THE WORLD

India (Developed 500 BCE – 600 CE) → Arab Scholars (Translated Indian texts) → Europe (Adopted as Hindu–Arabic Numerals) → Modern World (Universal Number System)

> "The numerals which are now common everywhere have their origin from India."
>
> – George Ifrah (Author, The Universal History of Numbers)

IMPACT ON THE WORLD

- Made mathematics universal and easy to learn.
- Enabled science, engineering, astronomy and technology.
- Became the backbone of modern computing and digital world.
- Empowered trade, finance and daily life calculations.
- One of India's greatest contributions to humanity.

EXAMPLES SHOWING THE POWER OF OUR NUMBER SYSTEM

PLACE VALUE

Number:	2	3	4	5
	Thousands	Hundreds	Tens	Ones
	↓	↓	↓	↓
	2000	300	40	5

Total = 2345

WITHOUT ZERO

Try writing 1005 without zero.
It would be impossible to show that there is 'no tens'.

ZERO makes the system complete and logical.

LARGE NUMBERS

10-digit number:
9,876,543,210

Easy to write, easy to read, easy to calculate!

POWER OF CALCULATION

$(125 \times 48) + (360 \div 15) - 29$
$= 6000 + 24 - 29$
$= 5995$

Efficient because of our positional system.

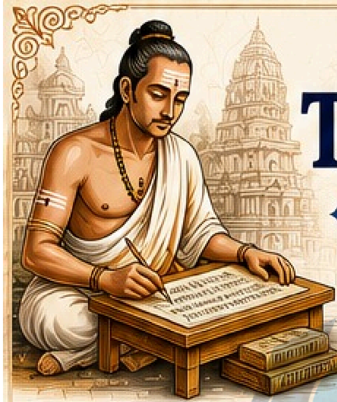
OUR HERITAGE, OUR PRIDE

The number system in India is not just a mathematical invention, it is a timeless intellectual legacy that continues to power the modern world.

From ancient wisdom to modern innovation – Indian numbers lead the way.

यत्र संख्याः, तत्र सिद्धिः ।
Where there are numbers, there is achievement.

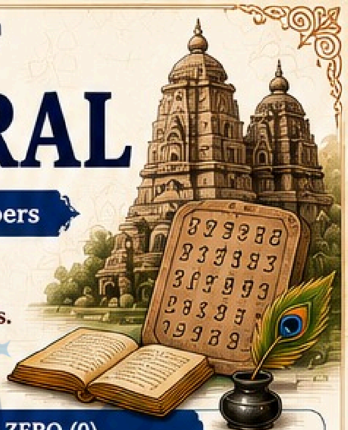
India counted the world into the future.



SALIENT FEATURES OF THE INDIAN NUMERAL

India's Gift to the World – A System that Counts Beyond Numbers

From ancient wisdom to global legacy –
The Indian numeral system is the foundation of modern mathematics.



1. DECIMAL (BASE-10) SYSTEM

India developed a positional decimal system with base 10.

Thousands	Hundreds	Tens	Ones
2	3	4	5
2000	300	40	5

The value of a digit depends on its position.

10 Base

2. USE OF ZERO (0)

- India is the only ancient civilization to conceptualize Zero as a number.
- Zero acts as a placeholder and enables representation of any large number.

0 शून्य (Zero)

3. PLACE VALUE SYSTEM

The value of a digit is determined by its place in the number.

Example: 5 2 7 8

5000 + 200 + 70 + 8

= 5,278

4. SYMBOLS FOR ALL DIGITS (0–9)

India developed unique symbols for all ten digits.

0	1	2	3	4	5	6	7	8	9
०	१	२	३	४	५	६	७	८	९

A complete set of digits – simple, logical and efficient.

5. POSITIONAL NOTATION

The same digit has different values at different positions.

Example:

Hundreds	Tens	Ones
2	2	2

200 + 20 + 2 = 222

6. WRITTEN FROM LEFT TO RIGHT

Indian numerals are written from left to right, which is natural and easy to read.

1 2 3 4 5 6 7 8 9

7. COMPACT AND CONCISE

Large numbers can be written in a short and compact form.

Example:

98,76,54,321

(Ten-digit number in just 10 symbols)

8. EASY TO PERFORM ARITHMETIC OPERATIONS

Addition, subtraction, multiplication and division become easier and systematic due to place value and zero.

+ − × ÷

9. ABILITY TO REPRESENT VERY LARGE AND VERY SMALL NUMBERS

Indian system can represent numbers of any size using the same 10 symbols.

Very Large:

7,89,00,00,00,00,00,00,000

(septendecillion)

∞ No Limit

Very Small:

0.0000000000000001 (10^{-16})

and beyond...

10. UNIVERSAL AND TIMELESS

- Developed in India, adopted all over the world.
- Used in mathematics, science, engineering, commerce, computing and daily life.
- A system that is simple, efficient and future-ready.

11. LOGICAL AND SYSTEMATIC STRUCTURE

Built on a strong logic:

10 Digits → Place Value → Zero → Infinite Numbers

This structure is the backbone of modern mathematics and computing.

12. CULTURAL AND HISTORICAL SIGNIFICANCE

- Described in ancient Indian texts like Sulba Sutras, Aryabhatiya, Brahmasphutasiddhanta, Lilavati.
- A remarkable achievement of Indian mathematicians.
- A proud contribution of India to the world.

EVOLUTION OF INDIAN NUMERALS

Brahmi (3rd century BCE)	Gupta (4th–6th century CE)	Siddham (7th–9th century CE)	Nagari (10th–12th century CE)	Modern Devanagari (13th century CE) onwards)	International Form (Used Worldwide)
[illegible]	१ २ ३ ४ ५	१ २ ३ ४ ५	१ २ ३ ४ ५	० १ २ ३ ४ ५ ६ ७ ८ ९	0 1 2 3 4 5 6 7 8 9

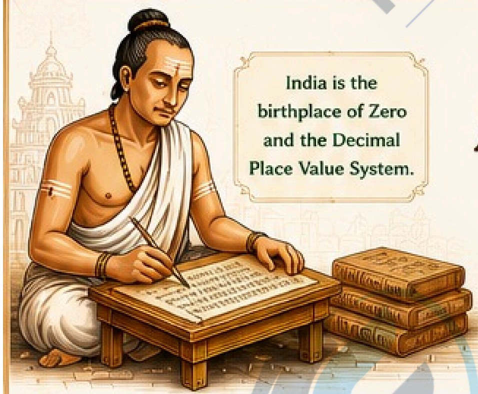
OUR HERITAGE, OUR PRIDE

The Indian numeral system is not just a way of counting, it is a brilliant invention that changed the world forever.

> Where there are numbers, there is order; where there is order, there is progress.

INDIAN NUMERALS –

INVENTED IN INDIA, PERFECTED FOR THE WORLD, POWERING THE FUTURE.



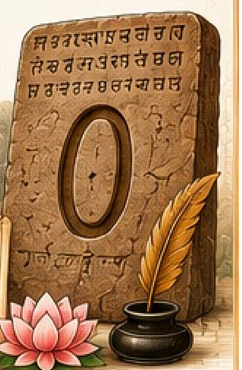
India is the birthplace of Zero and the Decimal Place Value System.

CONCEPT OF ZERO AND ITS IMPORTANCE

A SMALL SYMBOL THAT CHANGED THE WORLD

" Zero is not nothing,
Zero is something that makes everything possible.

The world adopted it, and humanity moved from counting to calculating, from limits to infinite possibilities.



1. HISTORICAL ORIGIN

- The concept of Zero was developed in India.
- Earliest use of a symbol for Zero found in Bakhshali Manuscript (3rd–4th century CE).
- Aryabhata (499 CE) used '0' in mathematical calculations.
- Brahmagupta (628 CE) gave rules for Zero in his book *Brahmasphutasiddhanta*.
- From India, Zero traveled to the Arab world, then to Europe, changing the course of mathematics and science.

Bakhshali Manuscript | Aryahbata | Brahmagupta

2. WHAT IS ZERO?

Zero (0) is a number that represents 'no quantity' or 'nothing', but it has a value and a place.

0

- Not negative
- Not positive
- It is neutral

Zero is neither less nor greater than any number.

3. PROPERTIES OF ZERO

✦ Addition	$a + 0 = a$
✦ Subtraction	$a - 0 = a, \quad 0 - a = -a$
✦ Multiplication	$a \times 0 = 0$
✦ Division	$0 \div a = 0 \quad (a \neq 0)$ $a \div 0 =$ Undefined
✦ Place Value	Zero as a placeholder gives value to other digits.

4. IMPORTANCE OF ZERO

1. FOUNDATION OF MATHEMATICS — Zero made the Decimal Place Value System possible, simplifying calculations.

2. EFFICIENCY IN CALCULATIONS — Makes arithmetic operations easier and more systematic.

3. SCIENCE & TECHNOLOGY — Essential in physics, computer science, engineering and space exploration.

4. MODERN COMPUTING & DIGITAL WORLD — 1010 0101 — Binary system (0 and 1) powers all digital devices, computers and AI.

5. FINANCE & ECONOMICS — Zero is crucial in banking, accounting, statistics and financial modeling.

6. MEASUREMENT & PRECISION — Enables precision in measurements, instruments and scientific data.

7. ENDLESS POSSIBILITIES — From zero come infinite numbers, shapes, patterns and innovations.

5. EXAMPLES SHOWING THE POWER OF ZERO

PLACE VALUE EXAMPLE

2	0	5
Hundreds	Tens	Ones
↓	↓	↓
200	0	5

Without zero, we cannot represent numbers like 205, 1008, or 10,000 correctly.

LARGE NUMBERS

1,00,00,000 (One Crore)

= 1 followed by 7 zeros

1	0	0	0	0	0	0	0

Zero allows us to write and understand very large numbers easily.

COMPARISON

Without Zero: 135

With Zero: 105

Zero changes the value and meaning of numbers completely.

MATHEMATICAL EXPRESSION

$(125 \times 8) + (360 \div 15) - 29$
$= 1000 + 24 - 29$
$= 1024 - 29$
$= 995$

Zero makes complex calculations possible and accurate.

6. JOURNEY OF ZERO TO THE WORLD

India (Ancient Times)
Concept of Zero developed by Indian mathematicians.

Arab World (7th–9th Century)
Indian numbers and Zero reached Arab scholars.

Europe (12th Century)
Introduced to Europe by Fibonacci.

Modern World
Now Zero is universal, used everywhere in science, technology & daily life.

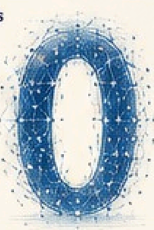
7. INSPIRING WORDS

" Zero is the greatest discovery of India. It is the basis of all modern mathematics and technology.
– Dr. Radhakrishnan

The introduction of Zero was a turning point in the intellectual history of mankind.
– George Ifrah

8. AMAZING FACTS

- Zero is the only number that is equal to its negative (0 = −0).
- Zero is a whole number, natural number and an integer.
- Zero has no Roman numeral – another reason why the Indian system is superior.
- A calculator or computer cannot work without zero!



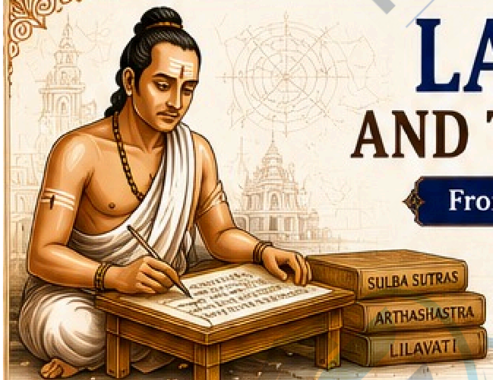
9. CONCLUSION

Zero is not "nothing".
It is "something" that gives value, meaning and power to numbers.
It is a symbol of human intelligence, creativity and progress.

From Zero, Infinite Possibilities!

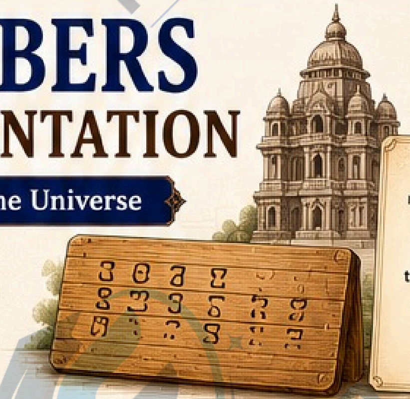
A SMALL SYMBOL FROM INDIA • A GIANT LEAP FOR HUMANITY

यत्र शून्यम्, तत्र पूर्णम् । – *Where there is Zero, there is Infinity.*



LARGE NUMBERS
AND THEIR REPRESENTATION

From Counting Seeds to Counting the Universe



> The Indian number system has no limits. It grows with the imagination and need of humanity.
>
> – Aryabhata

WHY DO WE NEED LARGE NUMBERS?

Astronomy	Population	Economics	Science	Computing
Distance between stars and galaxies	Counting billions of people	GDP, national debt, large transactions	Atoms, molecules and beyond	Data storage, IP addresses, etc.

INDIAN HERITAGE

- Indian mathematicians developed a powerful positional decimal system.
- They used large numbers for astronomy, geometry, commerce and time calculation.
- Texts like Aryabhatiya, Brahmasphutasiddhanta, Lilavati describe very large numbers.

1. THE INDIAN PLACE VALUE SYSTEM (BASE 10)

Crore (10^7)	Ten Lakh (10^6)	Lakh (10^5)	Ten Thousand (10^4)	Thousand (10^3)	Hundred (10^2)	Ten (10^1)	Ons (10^0)
1	2	3	4	5	6	7	8
1,00,00,000	10,00,000	1,00,000	10,000	1,000	100	10	1

India follows the International (Short) Scale with commas placed in groups of 2 digits after the last 3 digits.

Example: 12,34,56,789

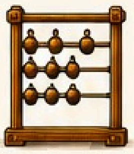
Crore Lakh Thousand Ones

2. INDIAN SYSTEM OF NAMING LARGE NUMBERS

Period	Number Name	Abbreviation	Value	In Digits	In Words
ONES	Ones	–	1	1	One
THOUSANDS	Thousand	K	10^3	1,000	One Thousand
LAKHS	Lakh	L	10^5	1,00,000	One Lakh
	Ten Lakh	TL	10^6	10,00,000	Ten Lakh
CRORES	Crore	Cr	10^7	1,00,00,000	One Crore
	Ten Crore	TC	10^8	10,00,00,000	Ten Crore
HIGHER ORDERS	Arab	A	10^9	1,00,00,00,000	One Arab
	Kharab	Kh	10^{11}	1,00,00,00,00,000	One Kharab
	Neel	N	10^{13}	1,00,00,00,00,00,000	One Neel
	Padma	P	10^{15}	1,00,00,00,000,00,000	One Padma
	Shankh	Sh	10^{17}	1,00,00,00,000,00,00,000	One Shankh
	Mahashankh	MSh	10^{19}	1,00,00,00,00,00,00,00,000	One Mahashankh

3. EXAMPLES OF LARGE NUMBERS

- **12,34,56,789**
  Twelve Crore Thirty Four Lakh Fifty Six Thousand Seven Hundred Eighty Nine
- **9,87,65,43,210**
  Nine Hundred Eighty Seven Crore Sixty Five Lakh Forty Three Thousand Two Hundred Ten
- **1,00,00,00,00,000,000**
  One Neel
- **7,89,00,00,00,00,00,000**
  Seven Padma Eighty Nine Shankh



Indian system can represent extremely large numbers with clarity and logic.

4. SCIENTIFIC (EXPONENTIAL) NOTATION

Very large numbers are often written in powers of 10.

General form: $N = a \times 10^n$ where $1 \leq a < 10$

Examples:

- 6,02,000,000 = 6.02×10^8
- 9,460,730,000,000 = 9.46073×10^{12}
- 1,00,00,00,00,00,000 = 1×10^{14}
- 3,00,00,00,00,00,00,00,000 = 3×10^{18}

5. COMPARING LARGE NUMBERS

Compare the number of digits. If equal, compare from left to right.

Compare: 98,76,54,321 and 1,02,34,567
Both have 8 digits. Compare from left.
Since $9 > 1$, 98,76,54,321 > 1,02,34,567

Compare: 1,23,45,67,890 and 1,23,45,67,899
Both have 10 digits. Compare from left.
Last digit $0 < 9$, 1,23,45,67,890 < 1,23,45,67,899

DID YOU KNOW?

The number of atoms in a grain of sand is about 10^{18}.

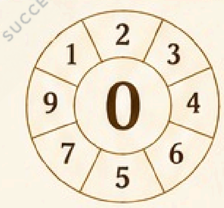
The number of stars in the observable universe is about 10^{22} to 10^{24}!

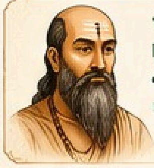
6. REAL LIFE REPRESENTATION OF LARGE NUMBERS

World Population (approx.)	8,000,000,000 (8 billion)
Distance from Earth to Sun	149,600,000 km (1.496×10^8 km)
Number of Cells in Human Body	37,200,000,000,000 (3.72×10^{13})
GDP of India (2023–24)	3,55,00,00,00,00,000 (3.55×10^{13} INR)
Age of Universe (approx.)	13,80,00,00,00,000,000,000 years (1.38×10^{10} years)

7. KEY FEATURES OF INDIAN NUMBER SYSTEM

- Positional – Value depends on the position of digit.
- Decimal – Based on 10 symbols (0–9).
- Efficient – Easy to write, read and calculate.
- Scalable – Can represent infinitely large numbers.
- Universal – Adopted and respected worldwide.





> "From the smallest unit to the largest cosmos, numbers connect everything. The Indian mind saw no limits in numbers."
>
> – Brahmagupta

KEY TAKEAWAY

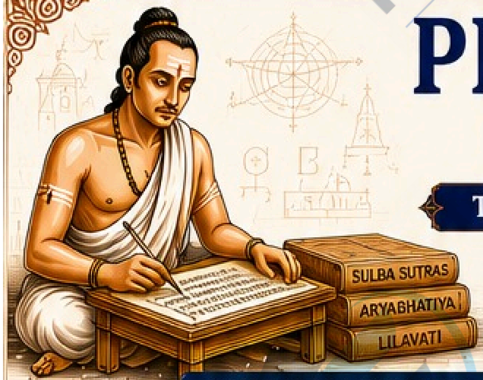
Large numbers help us understand the vastness of the world and beyond. The Indian numeral system gives us a powerful way to represent, compute and explore infinite possibilities.

गणितं विज्ञानानाम् मूलम् ।

Mathematics is the root of all sciences.

PLACE VALUE OF NUMERALS

The Position that Gives Value to a Digit



Indian mathematicians developed the place value system, the foundation of modern mathematics.

EXAMPLE

The digit 7 in 2 7 4 6

Thousands Place	Hundreds Place	Tens Place	Ones Place
= 7,000	= 700	= 70	= 7

WHAT IS PLACE VALUE?

In our number system, the value of a digit depends on its position (or place) in a number.

The same digit can have different values at different places.

THE INDIAN PLACE VALUE SYSTEM (BASE 10)

HIGHER ORDERS				LOWER ORDERS				
Ten Crore	Crore	Ten Lakh	Lakh	Ten Thousand	Thousand	Hundred	Ten	One
10^8	10^7	10^6	10^5	10^4	10^3	10^2	10^1	10^0
00,00,00,000	0,00,00,000	00,00,000	0,00,000	00,000	0,000	000	00	0
Crores		Lakhs		Thousands		Ones		

KEY POINTS

- We use 10 digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.
- Each place is 10 times the value of the place to its right.
- When we move one place to the left, we multiply by 10.
- When we move one place to the right, we divide by 10.

Place value makes our number system concise, efficient and powerful.

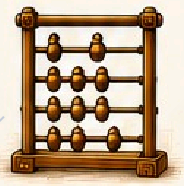
VALUE OF DIGITS AT DIFFERENT PLACES

EXAMPLE 1

Consider the number: 6 3 8 7 2

Digit	Place	Place Value	Expanded Form
6	Ten Thousands	$6 \times 10^4 = 60{,}000$	60,000 +
3	Thousands	$3 \times 10^3 = 3{,}000$	3,000 +
8	Hundreds	$8 \times 10^2 = 800$	800 +
7	Tens	$7 \times 10^1 = 70$	70 +
2	Ones	$2 \times 10^0 = 2$	2

Therefore, 63,872 = 60,000 + 3,000 + 800 + 70 + 2



Same digit, Different place, Different value.

EXAMPLE 2

Consider the number: 4 0 5 0 9

Digit	Place	Place Value	Expanded Form
4	Ten Thousands	$4 \times 10^4 = 40{,}000$	40,000 +
0	Thousands	$0 \times 10^3 = 0$	0 +
5	Hundreds	$5 \times 10^2 = 500$	500 +
0	Tens	$0 \times 10^1 = 0$	0 +
9	Ones	$9 \times 10^0 = 9$	9

Therefore, 40,509 = 40,000 + 0 + 500 + 0 + 9

EXAMPLE 3: SAME DIGIT, DIFFERENT PLACES

Let's see the digit 5 in different numbers.

- 5 in 5 $= 5 \times 10^0 = 5$
- 5 in 50 $= 5 \times 10^1 = 50$
- 5 in 500 $= 5 \times 10^2 = 500$
- 5 in 5,000 $= 5 \times 10^3 = 5{,}000$
- 5 in 50,000 $= 5 \times 10^4 = 50{,}000$

The value of 5 changes with its place (position).

EXAMPLE 4: PLACE VALUE IN INDIAN SYSTEM

Consider the number: 7,89,65,432

Crore	Lakh	Thousand	Ones
7	89	65	432

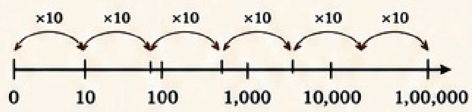
Expanded Form:

7,89,65,432 =

$7 \times 1{,}00{,}00{,}000 + 89 \times 1{,}00{,}000 + 65 \times 1{,}000 + 432$

= 7,00,00,000 + 89,00,000 + 65,000 + 432

EXAMPLE 5: PLACE VALUE ON NUMBER LINE



Each step to the left is 10 times bigger.
Each step to the right is 10 times smaller.

This is the power of place value!

IMPORTANCE OF PLACE VALUE

- It allows us to represent any large number using only 10 digits.
- It makes reading, writing and comparing numbers easy.
- It is essential for arithmetic operations.
- It is the foundation of our decimal number system and modern computing.

INTERESTING FACTS

- ★ The Indian place value system was one of the greatest contributions of India to the world.
- ★ It includes the concept of Zero (0), which acts as a placeholder.
- ★ Arab scholars learned this system from India and introduced it to the world.

QUICK RECAP

- Value depends on Place.
- Right → Smaller (÷ 10)
- Left → Bigger (× 10)
- Position gives Power to a Digit!

1 2 3 4 5 6 7 8 9 0

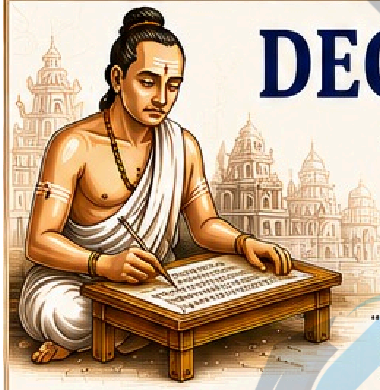
Where there is number, there is order;
where there is order, there is progress.

यत्र संख्या, तत्र सिद्धिः ।
(Where there are numbers, there is achievement.)

THE INDIAN PLACE VALUE SYSTEM – A TIMELESS GIFT TO HUMANITY

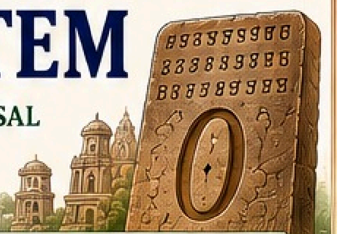
DECIMAL NUMBER SYSTEM

A SYSTEM BASED ON 10 – SIMPLE, POWERFUL AND UNIVERSAL



The decimal system is a number system in which numbers are represented using the digits 0 to 9 and each place value is a power of 10.

"Our Rishis gave the world the concept of Zero and the Decimal System."
– Aryabhata



THE GIFT OF INDIA TO THE WORLD

The concept of Zero (0) and the Decimal System were developed in ancient India.

1. WHAT IS THE DECIMAL NUMBER SYSTEM?

- It is a positional number system with base (or radix) 10.
- It uses ten digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.
- The value of a digit depends on its position (place).
- Each place to the left is 10 times greater.
- Each place to the right is 10 times smaller.

2. THE TEN DIGITS

0	1	2	3	4
5	6	7	8	9

These ten digits are enough to represent any number, big or small!

3. PLACE VALUE CHART

	Higher Orders			Whole Number Part		Decimal (Fractional) Part			Lower Orders	
	Thousands 10^3	Hundreds 10^2	Tens 10^1	Ones 10^0		Tenths 10^{-1}	Hundredths 10^{-2}	Thousandths 10^{-3}	Ten Thousandths 10^{-4}	...
Example:	2	3		5	.	6	4	7	9	...
	2,000	300		50	.	0.6	0.04	0.007	0.0009	...

4. PLACE VALUE OF A NUMBER – EXAMPLE

Consider the number: 4 2 7 . 3 8 6

Digit	Place	Place Value	Expanded Form
4	Hundreds	4×10^2	400
2	Tens	2×10^1	20
7	Ones	7×10^0	7
.	Decimal Point	–	
3	Tenths	3×10^{-1}	0.3
8	Hundredths	8×10^{-2}	0.08
6	Thousandths	6×10^{-3}	0.006

Expanded Form: 400 + 20 + 7 + 0.3 + 0.08 + 0.006 = 427.386

5. SALIENT FEATURES

- 10 — It is a base-10 system.
- 0-9 — It uses only ten digits (0 to 9).
- Positional System — The value of a digit depends on its position.
- 10× — Each place to the left is 10 times greater.
- 1/10 — Each place to the right is 1/10 (or 10 times) smaller.
- ∞ — It can represent very large as well as very small numbers.

6. EXAMPLES OF DECIMAL NUMBERS

- Whole Numbers: 0, 1, 25, 623, 10,000
- Decimal Numbers: 3.5, 0.25, 12.75, 98.006
- Very Small Numbers: 0.001, 0.0005
- Very Large Numbers: 1,234,567.89, 7,890,123,456.789

From counting grains to measuring the universe – the decimal system makes it possible!

7. CONVERSION USING PLACE VALUE

Example 1: Write 5,806.724 in expanded form

$5{,}806.724 = 5 \times 10^3 + 8 \times 10^2 + 0 \times 10^1 + 6 \times 10^0 + 7 \times 10^{-1} + 2 \times 10^{-2} + 4 \times 10^{-3}$

$= 5{,}000 + 800 + 0 + 6 + 0.7 + 0.02 + 0.004$

$= 5{,}806.724$

Example 2: Write in standard form

$3 \times 10^4 + 6 \times 10^2 + 2 \times 10^1 + 5 \times 10^{-1} + 3 \times 10^{-3}$

$= 30{,}000 + 600 + 20 + 0.5 + 0.003$

$= 30{,}620.503$

8. COMPARING DECIMAL NUMBERS

Compare: 72.45 and 72.405

Step 1: Equal number of decimal places

72.45 = 72.450

72.450 > 72.405

∴ 72.45 is greater than 72.405

Compare: 0.056 and 0.06

0.056 < 0.060

∴ 0.056 is smaller than 0.06

9. REAL-LIFE APPLICATIONS

- Money: ₹25.50, ₹100.75
- Measurement: 5.25 m, 0.75 kg
- Temperature: 36.6°C
- Science: 9.8 m/s² (acceleration due to gravity)
- Engineering: 0.005 mm precision
- Computers: Data storage, calculations

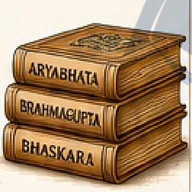
The decimal system is the language of science, technology and daily life.

10. ADVANTAGES OF THE DECIMAL SYSTEM

- Easy to understand and use.
- Simplifies arithmetic operations.
- Flexible for expressing fractions and large numbers.
- Essential for mathematics, commerce, science and technology.
- Universal acceptance.

11. A HISTORICAL GLORY

- The decimal system, along with Zero, was developed in India.
- Indian mathematicians like Aryahbata, Brahmagupta and Bhaskara made it powerful.
- It was later adopted by the Arab world and then by the entire globe.



12. INTERESTING FACTS

- There are infinite numbers in the decimal system.
- Between any two decimal numbers, there are infinite decimal numbers.
- The decimal system is the foundation of modern computing.

JOURNEY OF THE DECIMAL SYSTEM

Ancient India (2000 BCE – 5th Century CE)
Development of place value system and concept of Zero.

→ **Arab Mathematicians** (7th – 10th Century CE)
Spread the decimal system across the world.

→ **Europe** (12th – 16th Century CE)
Adopted as the Hindu-Arabic numeral system.

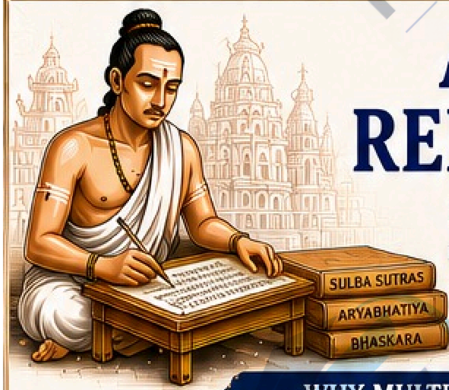
Modern World (17th Century onwards)
Universal use in science, commerce, technology and daily life.

SUMMARY

The Decimal Number System is a positional system with base 10 using digits 0–9. It is simple yet powerful, making it the most widely used number system in the world today.

A small idea that changed the world – Zero and Decimal!

" Where there are numbers, there is order; where there is order, there is progress."



APPROACHES TO REPRESENT NUMBERS

Numbers Can Be Represented in Many Ways

Different representations help us understand, compute and apply numbers efficiently in various fields.

> The diversity of number representations is the foundation of mathematics, science and technology.
>
> – Ancient Indian Wisdom

WHY MULTIPLE REPRESENTATIONS?

- Improve Understanding
- Simplify Calculations
- Reveal Different Properties
- Useful in Different Applications
- Universal & Timeless

EXAMPLE NUMBER

345

Let's see how this number can be represented in different ways.

MAJOR APPROACHES TO REPRESENT NUMBERS

1. POSITIONAL (PLACE VALUE) REPRESENTATION

Hundreds (10^2)	Tens (10^1)	Ones (10^0)
3	4	5
↓	↓	↓
300	40	5

$3 \times 10^2 + 4 \times 10^1 + 5 \times 10^0 = 345$

- Most common system (Base 10).
- Value depends on the position of the digit.
- Used in daily life (counting, money, measurements etc.).

2. EXPANDED FORM

300 + 40 + 5

$345 = 300 + 40 + 5$

- Shows the value of each digit separately.
- Helps in understanding place value.
- Useful in addition, subtraction and teaching.

3. WORD FORM

"Three Hundred Forty Five"

- Numbers written in words.
- Useful in banking, cheques, legal documents and readability.

4. ROMAN NUMERAL REPRESENTATION

CCCLXLV

Symbol	CCC	XL	V
Value	300	40	5
Total	345		

- Uses letters: I, V, X, L, C, D, M.
- Used in ancient times, clocks, book chapters, etc.
- 345 = CCCLXLV

5. BINARY REPRESENTATION (BASE 2)

2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
256	128	64	32	16	8	4	2	1
1	0	1	0	1	1	0	0	1

$345_{10} = 101011001_2$

- Uses only two digits: 0 and 1.
- Fundamental in digital computers & electronics.
- Each position is a power of 2.
- $345 = 256 + 64 + 16 + 8 + 1$

6. OCTAL REPRESENTATION (BASE 8)

8^3	8^2	8^1	8^0
512	64	8	1
5	3	3	1

$345_{10} = 531_8$

- Uses digits: 0 to 7.
- Shorter form of binary.
- Used in early computers and digital systems.
- $345 = 5\times64 + 3\times8 + 1$

7. HEXADECIMAL REPRESENTATION (BASE 16)

16^2	16^1	16^0
256	16	1
1	5	9

A 0 F

$345_{10} = 159_{16}$

- Uses digits: 0–9 and letters: A–F (10–15).
- Compact and widely used in programming, color codes, memory addresses.
- $345 = 1\times256 + 5\times16 + 9$

8. NUMBER LINE REPRESENTATION

345

300 310 320 330 340 350 360 370 380 380

- Shows position of a number on a line.
- Helps in understanding order, comparison, distance and estimation.
- 345 lies between 340 and 350.

9. PICTORIAL / VISUAL REPRESENTATION

100 100 100

10 10 10 10

1 1 1 1 1

300 + 40 + 5 = 345

- Uses pictures, blocks, dots or objects.
- Best for beginners and visual learners.
- Makes abstract numbers more concrete.

10. FRACTIONS & DECIMAL REPRESENTATION

Fraction Form	Decimal Form
$345 = \frac{345}{1}$	$345 = 345.0$

- Extends whole numbers to fractions and decimals.
- Essential in real life, science, engineering, finance, etc.

11. SCIENTIFIC NOTATION

$345 = 3.45 \times 10^2$

- Expresses numbers as a product of a number between 1 and 10 and a power of 10.
- Used for very large or very small numbers.
- Example:

$3450000 = 3.45 \times 10^6$

$0.00345 = 3.45 \times 10^{-3}$

12. COMPUTER & PROGRAMMING REPRESENTATIONS

- ASCII (Character Code) 'A' = 65
- Unicode 'A' = U+0041
- Boolean True = 1, False = 0

```
int num = 345;
bin: 101011001
hex: 0x159
oct: 0531
```

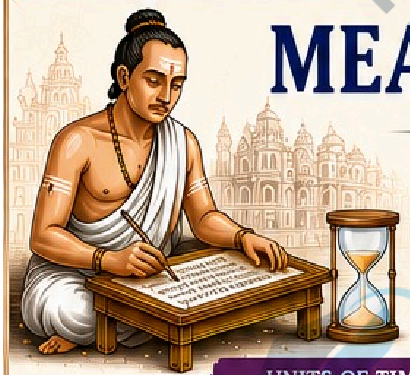
- Computers store and process numbers in various forms for efficiency and functionality.

One Number, Many Representations, Infinite Possibilities.

Different Approaches – Same Number – Deeper Understanding

Understanding representations is the key to mastering numbers and solving real-world problems.

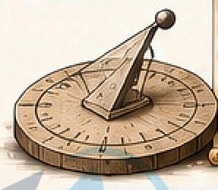
Unit - IV



MEASUREMENTS FOR TIME

→ Measuring Time, Understanding Our World

Time is the most valuable thing a man can spend.
– Theophrastus



From ancient sundials to atomic clocks, humanity has measured time to understand, plan and progress.

UNITS OF TIME – FROM SMALL TO LARGE

Second (s)	Minute (min)	Hour (h)	Day	Week	Month	Year	Decade	Century	Millennium
1 s	60 s	60 min	24 h	7 days	≈ 30 days	365 days (365.25)	10 years	100 years	1,000 years

WHY MEASURE TIME?

- Helps in planning daily activities
- Essential for science, technology and communication
- Needed for navigation, calendars and schedules
- Important in astronomy, history and research

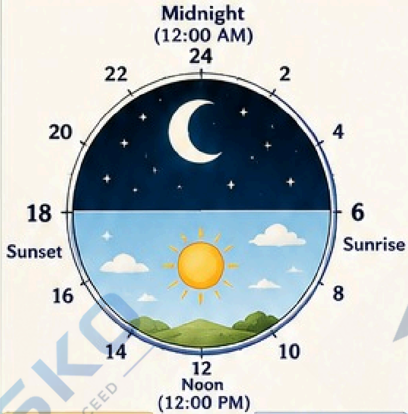
1. BASIC TIME CONVERSIONS

1 minute (min)	=	60 seconds (s)
1 hour (h)	=	60 minutes
1 day	=	24 hours
1 week	=	7 days
1 month	≈	30 days (28–31 days)
1 year	=	365 days (365.25 days)
1 leap year	=	366 days
1 decade	=	10 years
1 century	=	100 years
1 millennium	=	1,000 years

A leap year occurs every 4 years to keep our calendar in sync with the Earth's revolution around the Sun.

FEBRUARY 29

2. TIME IN A DAY



AM (Ante Meridiem) 12:00 AM to 11:59 AM

PM (Post Meridiem) 12:00 PM to 11:59 PM

3. ANCIENT INDIAN TIME MEASUREMENT

India has a rich tradition of time measurement used in astronomy, rituals and daily life.

Small Units (Traditional)

- 1 Truti ≈ 1/33,750 second
- 1 Lava = 3 Truti
- 1 Nimesha = 16 Lava
- 1 Kshana = 15 Nimesha
- 1 Kala ≈ 30 Kshana
- 1 Muhurta = 2 Kala (≈ 48 minutes)

Larger Units

- 1 Day (Ahoratra) = 30 Muhurta
- 15 Days = 1 Paksha
- 30 Days = 1 Month (Maas)
- 6 Months = 1 Ritu
- 12 Months = 1 Year

These units were used in Panchang (Hindu calendar), Vedic rituals, yoga, astrology and astronomy.

4. ASTRONOMICAL BASIS OF TIME

Rotation of Earth
One rotation on its axis
= 1 day (24 hours)

Revolution of Earth
One revolution around the Sun
= 1 year (≈ 365.25 days)

Phases of the Moon
Basis for months
(Lunar month ≈ 29.53 days)

5. MODERN PRECISE TIME MEASUREMENT

Quartz Clock – Uses crystal vibrations to keep accurate time. Accuracy: ±15 seconds/month

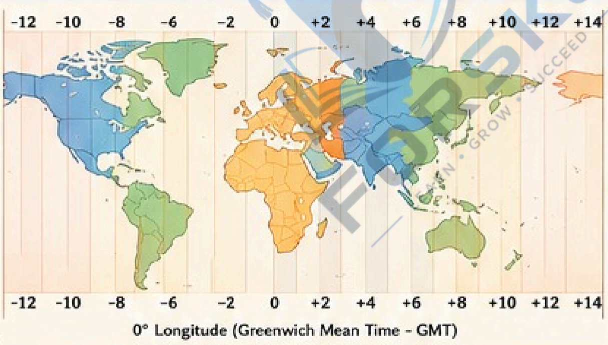
Atomic Clock – Uses vibrations of atoms (Cesium-133) for timekeeping. Accuracy: ±1 second in millions of years

GPS Time – Used in navigation, communication and global systems. Highly accurate and synchronized worldwide.

6. REAL-LIFE APPLICATIONS

- Transportation: Train, flight and bus schedules
- Communication: Time zones, internet sync
- Science: Experiments, research and space missions
- Health: Medicine timings, appointments
- Business: Meetings, deadlines and productivity
- Daily Life: Wake up time, study, exercise, rest

7. TIME ZONES OF THE WORLD



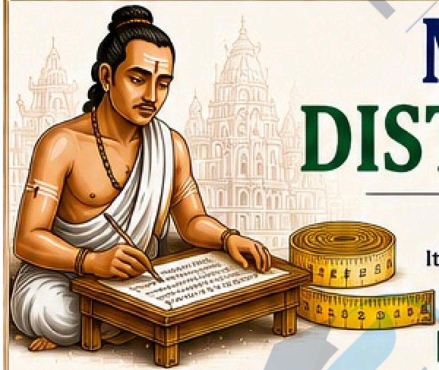
Earth is divided into 24 time zones.
Each time zone = 15° of longitude
= 1 hour difference

When it is 12:00 PM (Noon) at GMT (0°), time changes by ±1 hour for every time zone away from GMT.

QUICK FACTS

- Light takes 1.255 milliseconds to travel from the Earth to the Moon.
- The Universe is about 13.8 billion years old.
- The shortest time ever measured is the Planck Time: 5.39×10^{-44} seconds.
- Time is not constant – it can change with speed (Einstein's Theory of Relativity).

"Time is what we want most, but what we use worst."
– William Penn



MEASUREMENTS
DISTANCE AND WEIGHT

Measurement makes everything precise.
It helps us compare, calculate and understand the world around us.

PART 1: MEASUREMENTS OF DISTANCE (LENGTH)

1. COMMON UNITS OF LENGTH

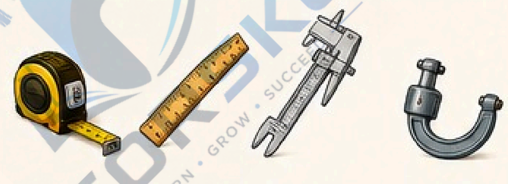
Unit	Symbol	In Terms of Meter	Example
Kilometer	km	1 km = 1,000 m	Distance between two cities
Meter	m	1 m = 1 m	Height of a door
Centimeter	cm	1 cm = 0.01 m	Length of a pen
Millimeter	mm	1 mm = 0.001 m	Thickness of a coin
Micrometer	µm	1 µm = 0.000001 m	Diameter of a human hair
Nanometer	nm	1 nm = 0.000000001 m	Size of a virus

2. CONVERSION OF LENGTH

- 1 km = 1,000 m
- 1 m = 100 cm
- 1 cm = 10 mm
- 1 mm = 1,000 µm
- 1 µm = 1,000 nm

Example:
2.5 km = 2.5 × 1,000 m
= 2,500 m
= 2,50,000 cm

3. MEASURING TOOLS



Measuring Tape (For long distances) | Meter Scale (For small lengths) | Vernier Caliper (For precise measurement) | Micrometer Screw Gauge (For very small lengths)

4. REAL LIFE EXAMPLES

- Road length between two cities is measured in kilometers.
- Height of a room is measured in meters.
- Length of a pencil is measured in centimeters.
- Size of bacteria is measured in micrometers or nanometers.

5. RELATION BETWEEN UNITS

km → × 1,000 → m → × 100 → cm → × 10 → mm → × 1,000 → µm → × 1,000 → nm

Move Right to Smaller Unit (Multiply)

Move Left to Larger Unit (Divide)

6. INTERESTING FACTS – DISTANCE

- Circumference of Earth ≈ 40,075 km
- Average distance from Earth to Moon ≈ 384,400 km
- Distance from Earth to Sun ≈ 150 million km
- An atom is about 0.1 nanometer (0.0000000001 m) in size!

PART 2: MEASUREMENTS OF WEIGHT (MASS)

1. COMMON UNITS OF WEIGHT (MASS)

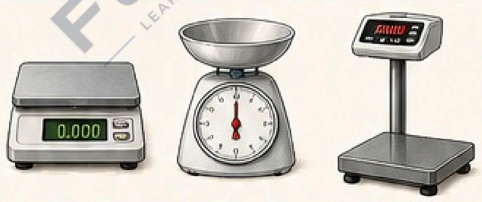
Unit	Symbol	In Terms of Kilogram	Example
Tonne (Metric Ton)	t	1 t = 1,000 kg	Weight of a car
Kilogram	kg	1 kg = 1 kg	Weight of a bag of rice
Gram	g	1 g = 0.001 kg	Weight of an apple
Milligram	mg	1 mg = 0.000001 kg	Weight of a medicine tablet

2. CONVERSION OF WEIGHT

- 1 t = 1,000 kg
- 1 kg = 1,000 g
- 1 g = 1,000 mg

Example:
3.2 kg = 3.2 × 1,000 g
= 3,200 g
= 3,200,000 mg

3. MEASURING TOOLS



Electronic Balance (For small mass) | Kitchen Scale (For daily use) | Platform Scale (For heavy load)

4. REAL LIFE EXAMPLES

- Goods in a truck are measured in tonnes.
- Sugar, rice, and flour are measured in kilograms.
- Fruits and vegetables are measured in grams.
- Medicines are measured in milligrams.

5. RELATION BETWEEN UNITS

t → × 1,000 → kg → × 1,000 → g → × 1,000 → mg

Move Right to Smaller Unit (Multiply)

Move Left to Larger Unit (Divide)

6. INTERESTING FACTS – WEIGHT

- Mass of Earth ≈ 5.97×10^{24} kg
- 1 liter of water has a mass of about 1 kilogram.
- A 1 rupee coin weighs about 3.85 grams.
- A typical tablet medicine weighs about 500 milligrams.

KEY POINTS

- Distance tells us how far an object is.
- Weight (mass) tells us how much matter an object contains.
- Different units are used for different needs.
- Standard units make communication and calculations easy.

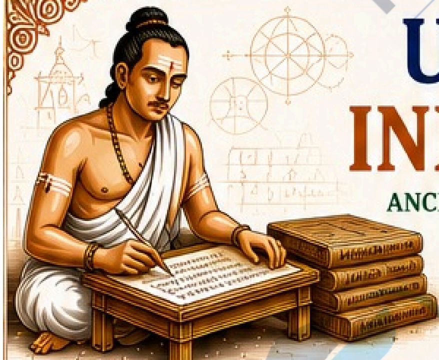
STANDARD UNITS (SI)

For Distance (Length)	For Weight (Mass)
Meter (m)	Kilogram (kg)

REMEMBER

"Measure everything measurable, make everything precise."

MEASUREMENT IS THE FIRST STEP TOWARDS KNOWLEDGE AND INNOVATION.



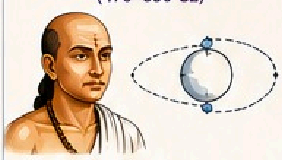
UNIQUE ASPECTS OF INDIAN MATHEMATICS

ANCIENT WISDOM • LOGIC • INNOVATION • UNIVERSAL IMPACT

> "From the Vedas to the modern world, Indian Mathematics has been a source of timeless knowledge and global inspiration."

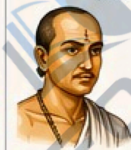
GREAT INDIAN MATHEMATICIANS AND THEIR CONTRIBUTIONS

ĀRYABHATA
(476–550 CE)



Author of Āryabhaṭīya. Introduced place value system and the concept of Zero.

BRAHMAGUPTA
(598–668 CE)



$$a + 0 = a$$
$$a - 0 = a$$
$$a \times 0 = 0$$

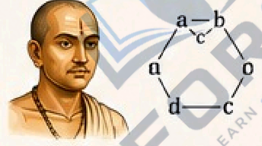
First to give rules for Zero and negative numbers in Brahmasphuṭasiddhānta.

BHĀSKARA II
(1114–1185 CE)

$$\frac{d(y)}{dx}$$

Wrote Siddhānta Śiromaṇi. Worked on calculus ideas, series expansions and advanced equations.

MAHĀVĪRA
(850–910 CE)



Early work on permutations and combinations in Gaṇita-sāra-sañgraha.

ŚRĪDHARA
(9th Century CE)

$$ax^2 + bx + c = 0$$

Gave method for solving quadratic equations in Bijagaṇita.

DISTINCTIVE FEATURES

- **Holistic Approach:** Mathematics was developed along with astronomy, philosophy and practical life.
- **Abstract Thinking:** Development of concepts like Zero, infinity, and large numbers.
- **Generalization:** Formulation of general rules and algorithms.
- **Systematic Methods:** Step-by-step procedures (algorithms) for calculations.
- **Integration with Real Life:** Used in architecture, trade, astronomy, music and rituals.
- **Strong Logical Foundations:** Emphasis on proofs, reasoning and sutra style of expression.

REVOLUTIONARY CONTRIBUTIONS

1. NUMBER SYSTEM

Place value system (Base 10)

3	2	7	6
Thousands	Hundreds	Tens	Ones

- Use of Zero as a number and a placeholder.
- Easy representation of large numbers.

0

The concept of Zero (śūnya) is India's gift to the world!

2. ALGEBRA (BĪJAGAṆITA)

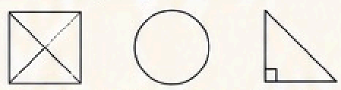
General equations and methods for solving them.

$$ax^2 + bx + c = 0$$

- Rules for operations with positives, negatives and Zero.
- Solutions of linear and quadratic equations.

3. GEOMETRY (ŚULBA SŪTRAS)

Precise geometric constructions for altars, temples and rituals.



- Accurate calculation of areas and lengths.
- Approximation of $\sqrt{2}$ (Baudhāyana's value).

4. TRIGONOMETRY (JYOTIṢA)

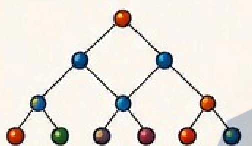
Introduction of trigonometric ratios for astronomical calculations.

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

- Use of sine tables (jya tables).
- Accurate prediction of eclipses and planets.

5. COMBINATORICS

Early study of permutations and combinations.



Number of ways to arrange 3 different objects taken 2 at a time = ${}^3P_2 = 6$

6. INFINITE SERIES

Work on infinite series and convergence.

$$1 + x + x^2 + x^3 + \cdots = \frac{1}{1-x}$$
$$(|x| < 1)$$

- Discussed in works of Bhāskara II.
- Ideas similar to modern calculus.

7. DECIMAL FRACTIONS

Use of decimal fractions and rules for operations.

$$\frac{1}{7} = 0.142857142857\cdots$$

(Repeating decimal)

- Clear rules for arithmetic operations.
- Seen in works of Nīlakaṇtha Somaayāji.

8. PRACTICAL APPLICATIONS

Indian Mathematics was applied in:

- Astronomy (planetary motions, eclipses, calendars)
- Architecture (temples, town planning, measurements)
- Trade & Commerce (weights, measures, interest calculations)
- Music (rhythmic patterns and combinations)

9. INDIAN MATHEMATICS AND THE WORLD

Indian mathematical ideas spread to the world through Arab scholars.

They translated Sanskrit texts and carried this knowledge to Europe.

This formed the foundation of modern mathematics.

DID YOU KNOW?

- The number system we use today is called the Hindu–Arabic Numeral System.
- Zero, place value, algebra, trigonometry, and combinatorics – all have deep roots in Indian Mathematics.
- Many concepts discovered in India centuries before the modern world!

THE TIMELESS LEGACY

Indian Mathematics is not just a collection of formulas, iet is a blend of wisdom, logic, creativity and practicality.

Think (Logically) → Discover (Patterns) → Solve (Problems) → Apply (In Life) → Share (Knowledge)

A HERITAGE THAT CONTINUES TO SHAPE OUR FUTURE

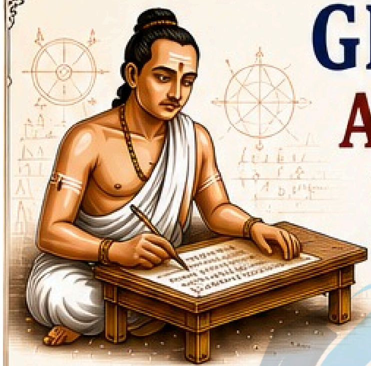
FAMOUS SUTRA STYLE

Indian mathematicians expressed complex ideas in short, precise sutras.

अल्पाक्षरं असंदिग्धं सारवद विश्वतोमुखम् ।
अस्त्यवे झाप्रियमाणं च सूत्रं सूत्रविदो विदुः ॥

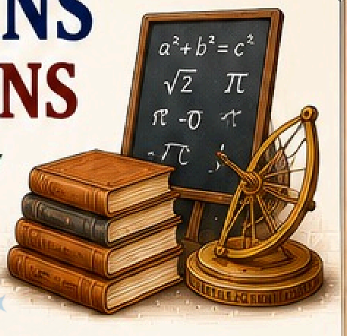
(A good sutra is short, unambiguous, essential, universal and pleasing.)

"India gave the world the science of numbers; let us value, learn and carry it forward."



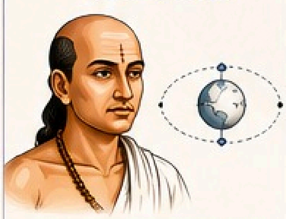
GREAT MATHEMATICIANS AND THEIR CONTRIBUTIONS

Brilliant Minds • Timeless Contributions • Eternal Legacy



> "Indian mathematicians have given the world ideas that are as infinite as numbers themselves."

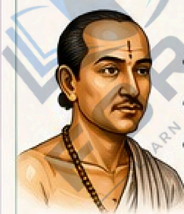
1 ĀRYABHATA
(476 – 550 CE)



Key Contributions

- Introduced the concept of Zero.
- Developed place value system.
- Calculated value of π (3.1416 approx).
- Explained Earth's rotation on its axis.
- Wrote Āryabhaṭīya.

2 BRAHMAGUPTA
(598 – 668 CE)



$a + 0 = a$

$a - 0 = a$

$a \times 0 = 0$

Key Contributions

- Gave rules for arithmetic with Zero.
- Introduced negative numbers.
- Solved linear and quadratic equations.
- Wrote Brahmasphuṭasiddhānta.

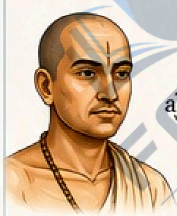
3 BHĀSKARA II
(1114 – 1185 CE)

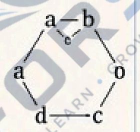
$\frac{d(y)}{dx}$

Key Contributions

- Worked on calculus (derivatives).
- Solved complex algebraic equations.
- Explained cyclic nature of the universe.
- Wrote Siddhānta Śiromaṇi.

4 MAHĀVĪRA
(850 – 910 CE)





Key Contributions

- Early work on permutations and combinations.
- Developed methods for mathematical reasoning.
- Wrote Gaṇita-sāra-saṅgraha.

5 ŚRĪDHARA
(9th Century CE)

$ax^2 + bx + c = 0$

Key Contributions

- Gave general method for solving quadratic equations.
- His work Bijagaṇita is a foundation of algebra.
- Used symbolic methods effectively.

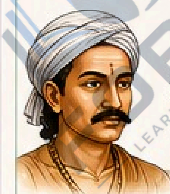
6 NĪLAKAṆTHA SOMAYĀJĪ
(1444 – 1544 CE)

$\frac{\sin x}{x}$

Key Contributions

- Developed infinite series for trigonometric functions.
- Accurately calculated π and astronomical values.
- Wrote Tantrasamgraha.

7 MADHAVA OF SAṄGAMAGRĀMA
(1340 – 1425 CE)

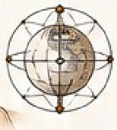


$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots$

Key Contributions

- Founder of Kerala School of Mathematics.
- Discovered infinite series for π, sin x, cos x, arctan x.
- Pioneered early calculus ideas.

8 VARĀHAMIHIRA
(505 – 587 CE)



Key Contributions

- Wrote Pañca-siddhāntikā on astronomy and mathematics.
- Improved calculation of eclipses and planetary positions.
- Bridged mathematics and astronomy.

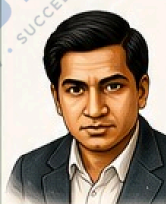
9 JYEṢṬHADEVA
(11th Century CE)

$\sqrt{2} = 1.41421356\ldots$ (approx)

Key Contributions

- Calculated $\sqrt{2}$ correct up to 11 decimal places.
- His accuracy was unmatched for his time.
- Example of deep numerical understanding.

10 S. RAMANUJAN
(1887 – 1920 CE)



Key Contributions

- Made extraordinary contributions to number theory, infinite series, continued fractions.
- Introduced many mathematical formulas and identities.
- His work continues to inspire modern mathematics.

COLLECTIVE IMPACT OF INDIAN MATHEMATICIANS

0 Introduced Zero and the Decimal System	Developed Algebra and Equations $ax^2 + bx + c = 0$	Advanced Trigonometry and Calculus	Discovered Infinite Series and Approximations $1 - \frac{1}{3} + \frac{1}{5} - \ldots$	Linked Mathematics with Astronomy and Practical Life	Built the Foundation for Modern Mathematics

DID YOU KNOW?

- The concept of Zero was born in India.
- Indian mathematicians calculated π correct to several decimal places centuries ago.
- Many ideas of calculus came from India long before Newton and Leibniz.
- The Kerala School was doing advanced mathematics in the 14th century.

A TIMELESS LEGACY

From ancient sages to modern geniuses, Indian mathematicians have shown that mathematics is not just numbers— it is creativity, logic and a way of life.

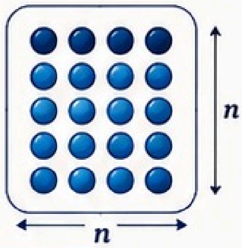
INSPIRATION FOR TOMORROW

Their curiosity, hard work and deep thinking continue to guide us.
Let us learn, explore and contribute to the beautiful world anf mathematics.

> "The glory of mathematics is that it is the most ancient of sciences and yet the most modern." — G.H. Hardy

ARITHMETIC – SQUARE OF NUMBER

The square of a number is the product of the number with itself.



1. DEFINITION

The square of a number n is written as n^2 and is calculated as:

$$n^2 = n \times n$$

Example:

$$5^2 = 5 \times 5 = 25$$

2. SQUARES OF NUMBERS (1 TO 20)

n	n^2	n	n^2	n	n^2
1	1	8	64	15	225
2	4	9	81	16	256
3	9	10	100	17	289
4	16	11	121	18	324
5	25	12	144	19	361
6	36	13	169	20	400
7	49	14	196		

3. PROPERTIES

- **+** Square of Positive Number is Positive. $(+n)^2 = n^2$
- **−** Square of Negative Number is Positive. $(-n)^2 = n^2$
- **1** Square of 1 is 1. $1^2 = 1$
- **0** Square of 0 is 0. $0^2 = 0$
- a^2 Square of a Number depends only on its magnitude, not on its sign.

4. PATTERNS IN SQUARES

Even Numbers Pattern

2^2	1	(Ends with 4)
4^2	16	(Ends with 6)
6^2	36	(Ends with 6)
8^2	64	(Ends with 4)
10^2	100	(Ends with 00)
12^2	144	(Ends with 44)

The unit digit of even squares circles: 4, 6, 6, 4, 0, 4, 6, 6, 4, 0...

Odd Numbers Pattern

1^2	1	(Ends with 1)
3^2	9	(Ends with 9)
5^2	25	(Ends with 5)
7^2	49	(Ends with 9)
9^2	81	(Ends with 1)
11^2	121	(Ends with 1)

The unit digit of odd squares circles: 1, 9, 5, 9, 1, 1, 9, 5, 9, 1...

Consecutive Odd Numbers Rule

The difference between the squares of consecutive numbers is an odd number.

$$(n+1)^2 - n^2 = 2n + 1$$

Examples:

$5^2 - 4^2 = 25 - 16 = 9 \quad (2\times4 + 1)$

$10^2 - 9^2 = 100 - 81 = 19 \quad (2\times9 + 1)$

5. SHORT TRICKS

Square of Numbers Ending with 5

If a number ends with 5, then:

$$(10a + 5)^2 = a(a+1)25$$

Examples:

$15^2 = (1 \times 2)25 = 225$

$35^2 = (3 \times 4)25 = 1225$

$95^2 = (9 \times 10)25 = 9025$

Square of Numbers Near Base (100)

$$(100 - a)^2 = 10000 - 200a + a^2$$

Examples:

$98^2 = (100 - 2)^2$
$= 10000 - 200\times2 + 2^2$
$= 9604$

$97^2 = (100 - 3)^2$
$= 10000 - 200\times3 + 3^2$
$= 9409$

Square of Numbers Ending with 1

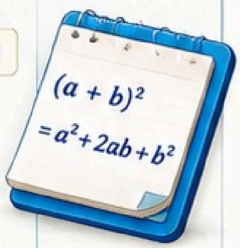
$$(10a + 1)^2 = a(a+2)01$$

Examples:

$11^2 = (1 \times 3)01 = 121$

$21^2 = (2 \times 4)01 = 441$

$31^2 = (3 \times 5)01 = 961$

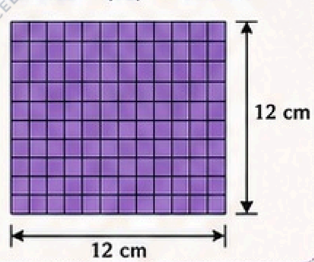


6. REAL-LIFE APPLICATIONS

- Area of a square with side n units is n^2 square units.
- Used in fields like architecture, engineering, and construction.
- Important in statistics, physics, computer graphics and more.

Example – Area of a Square

If the side of a square is 12 cm, then area = $(12)^2 = 144\ cm^2$



7. KEY TAKEAWAY

- ✅ Square of a number is the number multiplied by itself.
- ✅ Patterns help in quick calculation.
- ✅ Short tricks save time.
- ✅ Squares are useful in many real-life situations.

“Squares Build Strong Foundations in Mathematics and in Life.”

SQUARE ROOT

$\sqrt{x}$

The square root of a number is a value which, when **multiplied by itself**, gives the original number.

$(\sqrt{a})^2 = a$
$\sqrt{a} \geq 0$

1. DEFINITION

The square root of a non-negative number a is written as $\sqrt{a}$, such that $(\sqrt{a})^2 = a$ and $\sqrt{a} \geq 0$.

Example:

$\sqrt{25} = 5$, because $5^2 = 25$

$\sqrt{0} = 0$

Note:
Square root of a negative number is not real.

2. PERFECT SQUARES (1 TO 20)

n	n^2	n	n^2	n	n^2
1	1	8	64	15	225
2	4	9	81	16	256
3	9	10	100	17	289
4	16	11	121	18	324
5	25	12	144	19	361
6	36	13	169	20	400
7	49	14	196		

3. PROPERTIES

- $\sqrt{a} \geq 0$, for any real number $a \geq 0$
- $\sqrt{a^2} = |a|$
- $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$ (for $a \geq 0,\ b \geq 0$)
- $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$ $(b > 0)$
- $\sqrt{a^2 \times b} = a\sqrt{b}$ $(a \geq 0, b \geq 0)$
- $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$ $(a \geq 0,\ b > 0)$

4. METHODS OF FINDING SQUARE ROOT

1. Prime Factorization Method
2. Long Division Method

Example (Prime Factorization Method)

Find $\sqrt{72}$

$72 = 2 \times 2 \times 2 \times 3 \times 3$

$= 2^2 \times 3^2 \times 2$

$\sqrt{72} = \sqrt{2^2 \times 3^2 \times 2}$

$= 2 \times 3\sqrt{2} = 6\sqrt{2}$

5. APPROXIMATE VALUES

$\sqrt{2} \approx 1.414$ $\sqrt{3} \approx 1.732$ $\sqrt{5} \approx 2.236$

$\sqrt{6} \approx 2.449$ $\sqrt{7} \approx 2.646$ $\sqrt{10} \approx 3.162$

Useful in mental calculations and estimations.

6. SHORT TRICKS

- Numbers ending with even number of zeros:

$\sqrt{a \times 10^{2n}} = 10^n \sqrt{a}$

Example:

$\sqrt{3200} = \sqrt{32 \times 100} = 10\sqrt{32}$

$= 10 \times 4\sqrt{2} = 40\sqrt{2}$

- Numbers near perfect squares:

If $n^2 < a < (n+1)^2$, then

$n < \sqrt{a} < n + 1$

Example:

$25 < 30 < 36$

$5 < \sqrt{30} < 6$

7. REAL-LIFE APPLICATIONS

- Used in geometry (diagonals, distance)
- Physics (formulas involving distance, speed, energy)
- Engineering & construction
- Finance & statistics

SERIES AND PROGRESSIONS

A **series** is the sum of the terms of a sequence. A **progression** is a sequence in which the terms follow a specific pattern.

1. ARITHMETIC PROGRESSION (A.P.)

A progression in which the difference between consecutive terms is constant.

$a,\ a+d,\ a+2d,\ a+3d,\ \ldots,\ a+(n-1)d$

First term (a) Common difference (d) n^{th} term

n^{th} term $(t_n) = a + (n-1)d$

Sum of first n terms (S_n)

$= \frac{n}{2}[2a + (n-1)d]$

Example:

For A.P.: 2, 5, 8, 11, …

$a = 2,\ \ d = 3$

$t_5 = 2 + (5-1)\times 3 = 14$

$S_5 = \frac{5}{2}[2\times 2 + (5-1)\times 3] = \frac{5}{2}(4+12) = 40$

Key Points

- ✓ $d = t_2 - t_1 = t_3 - t_2$
- ✓ If $d > 0 \rightarrow$ A.P. is increasing
- ✓ If $d < 0 \rightarrow$ A.P. is decreasing

2. GEOMETRIC PROGRESSION (G.P.)

A progression in which the ratio between consecutive terms is constant.

$a,\ ar,\ ar^2,\ ar^3,\ \ldots,\ ar^{n-1}$

First term (a) Common ratio (r) n^{th} term

n^{th} term $(t_n) = ar^{n-1}$

Sum of first n terms (S_n)

If $r \neq 1$: $S_n = a\frac{1-r^n}{1-r}$

If $r = 1$: $S_n = na$

Example:

For G.P.: 3, 6, 12, 24, …

$a = 3,\ \ r = 2$

$t_5 = 3\times 2^{5-1} = 48$

$S_5 = 3\frac{1-2^5}{1-2} = 3\frac{(1-32)}{-1} = 93$

Key Points

- ✓ $r = \frac{t_2}{t_1} = \frac{t_3}{t_2}$
- ✓ If $|r| > 1 \rightarrow$ G.P. is increasing
- ✓ If $0 < |r| < 1 \rightarrow$ G.P. is decreasing

3. IMPORTANT FORMULAS

- Sum of first n natural numbers:

$1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$

- Sum of squares of first n natural numbers:

$1^2 + 2^2 + 3^2 + \ldots + n = \frac{n(n+1)(2n+1)}{6}$

- Sum of cubes of first n natural numbers:

$1^3 + 2^3 + 3^3 + \ldots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$

4. SPECIAL SERIES

- $1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$
- $2 + 4 + 6 + \ldots + 2n = n(n+1)$
- $1 + 3 + 5 + \ldots + (2n-1) = n^2$
- $1^2 + 2^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$
- $1^3 + 2^3 + \ldots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$

5. REAL-LIFE APPLICATIONS

A.P.: Used in savings plans, salary increments, construction steps, emissions data, etc.

G.P.: Used in population growth, compound interest, radioactive decay, loan repayments, etc.

Series: Used in statistics, physics, computer science, signal processing, and many more.

6. QUICK RECAP

Type	General Form	n^{th} Term	Sum of n Terms
A.P.	$a, a+d, a+2d, \ldots$	$a + (n-1)d$	$\frac{n}{2}[2a + (n-1)d]$
G.P.	$a, ar, ar^2, \ldots$	ar^{n-1}	$a\frac{1-r^n}{1-r}$ $(r \neq 1)$ na $(r = 1)$

"Mathematics is not about numbers, equations, computations, or algorithms: it is about understanding." — William Paul Thurston

1. VALUE OF PI (π)

Pi (π) is the constant ratio of the circumference of a circle to its diameter.

$$\pi = \frac{C}{d}$$

C = Circumference
d = Diameter

1. VALUE

π is an irrational, non-terminating non-repeating decimal.

$\pi \approx 3.14159265358979323846...$

Common Approximations:

$\pi \approx 3.14$ (good for general use)
$\pi = 22/7$ (traditional & useful)
$\pi \approx 355/113$ (more accurate)

2. KEY FORMULAS

Circumference $(C) = 2\pi r$
Area $(A) = \pi r^2$
Diameter $(d) = 2r$

Arc Length $(L) = (\theta/360^\circ) \times 2\pi r$
Area of Sector $= (\theta/360^\circ) \times \pi r^2$
where r = radius, θ = central angle

3. INTERESTING FACTS

- π has infinite digits without any pattern.
- It appears in many areas of mathematics, physics, engineering and nature.
- π is used in calculations involving circles, waves, probability, statistics, and more.
- Pi Day is celebrated on March 14 (3/14).

MAR 14 PI DAY

4. PLACE VALUE OF π (FIRST 20 DIGITS)

Digit	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
π	3.	1	4	1	5	9	2	6	5	3	5	8	9	7	3	2	3	8	4	6

$\pi = 3.14159265358979323846...$

5. REAL-LIFE APPLICATIONS

Engineering, Architecture, Physics, Computer Graphics, Navigation

2. ALGEBRA

$ax + b = c$

Algebra is the branch of mathematics that uses symbols and letters to represent numbers and relationships.

1. BASIC CONCEPTS

- A variable is a symbol (like x, y, z) representing a number.
- A constant is a fixed number.
- An expression is a combination of variables, constants and operations.
- An equation shows equality between two expressions.

Example:
Expression: $3x + 5$
Equation: $3x + 5 = 14$

2. ALGEBRAIC IDENTITIES

- $(a+b)^2 = a^2 + 2ab + b^2$
- $(a-b)^2 = a^2 - 2ab + b^2$
- $(a+b)(a-b) = a^2 - b^2$
- $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
- $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$
- $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

3. LINEAR EQUATION

General form:
$ax + b = 0 \quad (a \neq 0)$

Solution:
$x = -\frac{b}{a}$

Example:
$2x + 5 = 15$
$2x = 10$
$x = 5$

4. POLYNOMIALS

- A polynomial is an expression with one or more terms.

Example:
$P(x) = 3x^3 - 2x^2 + 5x - 7$

- Degree of Polynomial = Highest power of the variable (Here, degree = 3)

5. FACTORING

- Common Factor: $ax + ay = a(x + y)$
- Difference of Squares: $a^2 - b^2 = (a+b)(a-b)$
- Perfect Square Trinomial: $a^2 + 2ab + b^2 = (a+b)^2$
- Grouping: $ax + ay + bx + by = a(x+y) + b(x+y) = (a+b)(x+y)$

Example:
Factorize: $x^2 + 5x + 6$
$x^2 + 5x + 6 = x^2 + 2x + 3x + 6$
$= x(x+2) + 3(x+2)$
$= (x+2)(x+3)$

6. REAL-LIFE APPLICATIONS

- Engineering & Physics
- Economics
- Computer Science
- Everyday Problem Solving

KEY TAKEAWAY

Algebra helps us model, solve and understand real-world problems using symbols and equations.

3. BINARY MATHEMATICS

101101

0101

Binary mathematics uses only two digits: 0 and 1.
It is the foundation of digital systems and computers.

1. BINARY DIGITS

The binary number system (Base 2) uses:

0 , 1

Each position in a binary number represents a power of 2.

2. PLACE VALUE IN BINARY

Position	7	6	5	4	3	2	1	0
Power of 2	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Value	128	64	32	16	8	4	2	1

Example: $(1011011)_2$
$= 1 \times 64 + 0 \times 32 + 1 \times 16 + 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1$
$= 64 + 16 + 8 + 2 + 1 = 91_{10}$

3. BINARY ADDITION

$0 + 0 = 0$
$0 + 1 = 1$
$1 + 0 = 1$
$1 + 1 = 10$ (0, carry 1)

Example:
$$\begin{array}{r} 1011 \\ +\ 1101 \\ \hline 11000 \end{array}$$

4. BINARY SUBTRACTION

$0 - 0 = 0$
$1 - 0 = 1$
$1 - 1 = 0$
$10 - 1 = 1$ (borrow 1)

Example:
$$\begin{array}{r} 1101 \\ -\ 1011 \\ \hline 0010 \end{array}$$

5. BINARY MULTIPLICATION

$0 \times 0 = 0$
$0 \times 1 = 0$
$1 \times 0 = 0$
$1 \times 1 = 1$

Example:
$$\begin{array}{r} 101 \\ \times\ 11 \\ \hline 101 \\ 101 \\ \hline 1111 \end{array}$$

6. BINARY TO DECIMAL QUICK METHOD

Step 1: Write the place values (powers of 2).
Step 2: Multiply each bit by its place value.
Step 3: Add all the results.

Example: $(11010)_2$

16	8	4	2	1
1	1	0	1	0

$= 1 \times 16 + 1 \times 8 + 0 \times 4 + 1 \times 2 + 0 \times 1 = 26_{10}$

7. DECIMAL TO BINARY (DIVISION METHOD)

Step 1: Divide the number by 2.
Step 2: Write the remainder (0 or 1).
Step 3: Repeat with the quotient.
Step 4: Read remainders bottom to top.

Example: Convert $(25)_{10}$ to binary

$25 \div 2 = 12$ R1
$12 \div 2 = 6$ R0
$6 \div 2 = 3$ R0
$3 \div 2 = 1$ R1
$1 \div 2 = 0$ R1

$(25)_{10} = (11001)_2$

8. APPLICATIONS

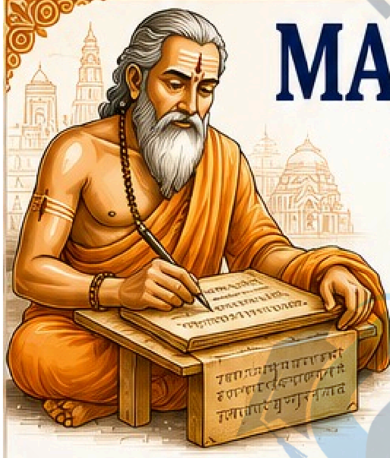
- Computers & Processors
- Digital Electronics
- Data Storage
- Networking & Security
- Programming

KEY TAKEAWAY

Binary mathematics is simple but powerful — it drives all digital technology around us.

Mathematics is the language of logic, patterns and possibilities.
From π to Algebra to Binary – numbers build the digital world!

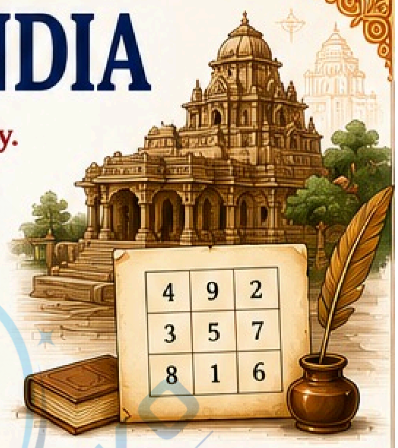
MAGIC SQUARES IN INDIA



Ancient Wisdom. Perfect Numbers. Timeless Legacy.

Magic Squares are special grids of numbers arranged so that the sum of numbers is the same in each row, column and diagonal.

> "From ancient inscriptions to modern mathematics, India has a rich tradition of understanding patterns, numbers and perfection."



1. ORIGIN AND HISTORY

- Magic Squares have been known in India for more than 2000 years.
- References are found in ancient texts like Atharva Veda, Brihat Samhita (by Varahamihira, 6th century CE) and ancient temples.
- They were used in astronomy, architecture, yantras, rituals and art.
- Indian mathematicians explored patterns, symmetry and numerical harmony.

यथा पिण्डे तथा ब्रह्माण्डे

(As in the body, so in the universe)

2. THE FAMOUS 3×3 MAGIC SQUARE

The earliest known magic square in India is the "Lo Shu Square" mentioned by Varahamihira.

4	9	2
3	5	7
8	1	6

Magic Constant = 15

Row sum = 15
Column sum = 15
Diagonal sum = 15

3. MAGIC SQUARES IN INDIAN TEMPLES AND ARCHITECTURE

- Magic Squares are carved on the walls, floors and ceilings of many temples.
- Believed to bring balance, positivity and protection.
- Examples:
- Konark Sun Temple (Odisha)
- Airavatesvara Temple (Tamil Nadu)
- Melkote Temple (Karnataka)
- Bhuleshwar Temple (Maharashtra) and many more.

16	3	2	13
5	10	11	8
9	6	7	12
4	15	14	1

4×4 Magic Square (Found in Indian Temples)

4. EXAMPLE OF 4×4 MAGIC SQUARE

16	3	2	13
5	10	11	8
9	6	7	12
4	15	14	1

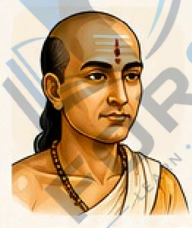
Magic Constant = 34

Row sum = 34
Column sum = 34
Diagonal sum = 34

WHY ARE THEY SPECIAL?

- Symbolize harmony and balance.
- Show mathematical beauty and symmetry.
- Connect mathematics with spirituality.
- Inspire patterns in art, music, astrology, and vastu shastra.

5. INDIAN MATHEMATICAL GENIUS



Indian mathematicians like Aryabhata, Brahmagupta, Bhaskara II and others studied number patterns deeply.

Their work influenced mathematics across the world.

6. TYPES OF MAGIC SQUARES

Odd Order
(3×3, 5×5, ...)

Examples:

4	9	2
3	5	7
8	1	6

Easy to construct. Always have a magic square.

Even Order
(4×4, 6×6, ...)

Examples:

16	3	2	13
5	10	11	8
9	6	7	12
4	15	14	1

More complex construction.

Singly Even Order
(6×6, 10×10, ...)

Need special methods.

35	1	6	26	19	24
3	32	7	21	23	25
31	9	2	22	27	20
8	28	33	17	10	15
30	5	34	12	14	16
4	36	29	13	18	11

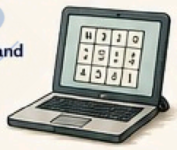
Doubly Even Order
(4×4, 8×8, ...)

Easy pattern methods.

16	3	2	13
5	10	11	8
9	6	7	12
4	15	14	1

7. MODERN APPLICATIONS

- Used in design, coding, puzzles and games.
- Helpful in teaching patterns and logic.
- Inspires algorithms and computer science.
- Still used in astrology, vastu and numerology.



DID YOU KNOW?

- The 3×3 Lo Shu Square is believed to be mystical and is associated with the planet Mercury.
- The numbers 1 to 9 can create only one 3×3 magic square (in different rotations).

A TIMELESS LEGACY

Magic Squares in India are not just about numbers, they are a perfect blend of mathematics, art, science and spirituality.

They represent the Indian vision of order, harmony and infinity.

TRY THIS!

Verify the magic constant for the 3×3 and 4×4 squares.

Can you create your own magic square?

START

Numbers are the language with which God has written the universe.
– Galileo Galilei